OptoTACH

A crystal-timed, microcontroller-free non-contact optical tachometer

Complete design, implementation and verification record of the KiCad project tach_v1.0

Item	Value
Project	tach_v1.0 (product name OptoTACH)
Document type	Technical report: design, analysis, layout, manufacturing, assembly, debugging, verification
Report revision / date	1.0 / 27 September 2026
EDA tool	KiCad 10.0.6 (Eeschema, Pcbnew, kicad-cli); all numbers re-extracted from the project files on 27 Sep 2026
Design state described	Schematic: wired A3 sheet (2026-09-23). PCB: routed, filled, 3D models attached (2026-09-23). Gerbers: fabrication set of 2026-09-17
Hardware state	Board 2 assembled and working; RPM readings close to a commercial tachometer (board 1 damaged during bring-up, Section 11)

Conventions used in this report

- **Source tags.** Every quantitative statement that is not self-evident carries a tag: **[F]** read from the design files, **[D]** manufacturer datasheet, **[C]** calculated in this report (formula shown), **[S]** simulated, **[O]** observed on the assembled hardware, **[A]** stated assumption. Nothing is presented as measured unless tagged [O].
- **Net names.** Nets are named as on the PCB (for example /BURST, /GATE_SEL). The redrawn schematic carries KiCad auto-names for the same nets (for example Net-(Q2-C)); the one-to-one mapping is Appendix B.
- **Supply.** V_{CC} = 6.0 V nominal (4 × AA alkaline) unless stated. The silkscreen marks the input PWR 5–6V.
- **Values.** Where the fitted part differs from the drawn value, both are given (Section 9.2). Calculations use the fitted value.
- **Coordinates.** PCB coordinates are KiCad absolute millimetres. The board outline spans X 99.975–200.025 mm, Y 49.975–150.025 mm.

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Abstract

OptoTACH is a hand-held, non-contact optical tachometer built only from CMOS logic, two NE555 timers and discrete parts: no microcontroller and no firmware. A TCRT5000 reflective sensor produces one pulse per revolution from a reflective mark on the rotor. A 32.768 kHz crystal divided by two CD4060 counters opens a counting window of exactly 1.000 s once every 4 s. In RPM mode every revolution is converted into a burst of exactly 60 counts by a latch-gated NE555 oscillator and a ÷10 × ÷6 CD4017 chain, so the four CD4026 decade counters display revolutions per minute directly; RPS and free-running counter modes use the raw pulses. The design is a rebuild of a published circuit whose CD40110 counters are not available in Bangladesh; replacing them with CD4026 (which has no display latch) required a new blank-while-counting display scheme implemented with the spare gates of the existing CD4093.

The circuit has 95 components (13 ICs) on a 100.05 × 100.05 mm, two-layer, all-through-hole PCB with 149 nets, 372 connected pins, 942 track segments (3,794.4 mm), 14 vias and a ground pour on both layers. ERC reports 0 violations and DRC 0 errors and 0 unconnected items; a pin-by-pin comparison proves the schematic and the copper describe identical connectivity. Analysis gives a resolution of 60 RPM (one count), a time-base error below 0.005 %, and a maximum speed of 4,809 RPM with C3 = 100 nF, above which the reading silently collapses to half; with C3 = 22 nF the limit rises to 21,859 RPM, beyond the 9 999 the display can show. Two design faults found during bring-up (a TCRT5000 symbol with the IR-LED anode and cathode swapped, and board damage from a supply short) and three design-margin findings (reset drive, power-on state of the ÷60 chain, simultaneous switch positions) are documented with root causes and corrections for a revision 1.1. The second assembled board works, and its readings of a PC case fan at 30 mm were close to those of a commercial Lutron DT-2236 tachometer [O].

Keywords: optical tachometer; TCRT5000; gated frequency counter; CD4026; CD4060; NE555; frequency multiplication; KiCad; design verification.

1 Introduction

1.1 Background

A tachometer measures rotational speed. Contact tachometers couple a wheel to the shaft; non-contact optical tachometers instead detect a mark on the rotor with reflected light and count the marks per unit time. The counting approach is attractive for a discrete design because its accuracy depends on a time base, which a quartz crystal provides with parts-per-million accuracy, rather than on analogue calibration.

A counter that is open for exactly one second and counts one pulse per revolution shows revolutions per second (RPS). To show revolutions per minute (RPM) directly from the same one-second window, each revolution must be turned into 60 counts. OptoTACH does this with a counting frequency multiplier: each revolution releases a burst from a free-running oscillator, and the burst is stopped by counting, not by timing, so the number of counts per revolution is exactly 60 regardless of the oscillator frequency.

1.2 Objectives

- Measure shaft or fan speed without contact and display RPM, RPS or a raw count on four 7-segment digits.
- Use only parts obtainable in Bangladesh; no microcontroller, no programming step.
- Put the design on a manufacturable two-layer PCB with hand-solderable through-hole parts and IC sockets.
- Verify the design before and after manufacture: electrical rules, design rules, connectivity, timing and accuracy.

1.3 Contributions of this build

- Substitution of CD40110 (counter with display latch) by CD4026 (no latch), with a 1 s-in-4 s gated count and display blanking made from gates already on the board (Section 2).
- A complete KiCad 10 implementation: custom CD4026 and TCRT5000 symbols, a custom TCRT5000 footprint, a routed and verified 100 × 100 mm board, a fully wired schematic (0 net labels), and a manufacturing package (Sections 6–8).

- Quantitative analysis: time base, ×60 burst timing, RPM ceiling and its sensitivity to C3, quantisation, sensor switching currents from datasheet thresholds, and a behavioural simulation over 1–100 Hz (Section 5).
- A staged bring-up procedure with test points and expected readings, and a fault log with root causes (Sections 10–11).
- An errata list and a concrete revision-1.1 change list (Sections 13–14).

2 Design origin and constraints

2.1 Reference design

The circuit topology follows the published project “Non Contact Tachometer Without Microcontroller” (Mousa Simple Projects, 2020). Its parts list [F: Tachometer/Parts list.txt] is reproduced in Table 1 for traceability. The reference uses four CD40110 up/down counter-latch-decoder-drivers.

Table 1. Parts list of the reference design [F].

Reference part	Qty	Reference part	Qty
IC 555	2	Capacitor 100 nF	5
IC 40110	4	Capacitor 100 µF	1
IC 4060	2	Capacitor 1 µF	1
IC 4017	2	Capacitor 40 pF	1
IC 4093	1	Resistor 1 kΩ	34
TCRT5000 sensor	1	Resistor 2 MΩ	2
DIP switch, 4 channel	1	Resistor 10 kΩ / 33 kΩ / 20 kΩ	1 / 1 / 1
Transistor 2N2222	2	Resistor 10 MΩ / 200 Ω / 220 kΩ	1 / 1 / 1
Optocoupler PC817	2	Crystal 32.768 kHz	1
7-segment CC display	4	LED; diode 1N4148; push button	1; 1; 1

2.2 Sourcing constraint and its consequence

The CD40110 is not sold by the local distributors, so tach_v1.0 uses four CD4026 decade counters with decoded 7-segment outputs. The CD4026 has no output latch: its segments follow the counter continuously, so a display driven by a counter that is counting is unreadable. The adopted solution (“option 3” in the project log) is:

- count for exactly 1 s once every 4 s: the window is the logical AND of the 0.5 Hz and 0.25 Hz outputs of the second CD4060 (Section 4.4);
- blank the display while counting through the CD4026 DISPLAY ENABLE input (pin 3), so the digits show a steady result for the remaining ~3 s;
- generate both gate signals with the two spare gates of the existing CD4093, so all four of its gates are used and the IC count is unchanged.

2.3 Differences from the reference design

Table 2. Design differences [F].

Item	Reference (parts list)	tach_v1.0	Reason
Counter / decoder	4 × CD40110	4 × CD4026 (U11–U14)	availability
Display hold	latch inside CD40110	blanking during a 1 s window every 4 s (U2C, U2D)	CD4026 has no latch
Segment resistors	1 kΩ (34 × 1 kΩ listed)	28 × 220 Ω (R20–R26, R30–R36, R40–R46, R50–R56)	CD4026 output is current-limited (Section 4.8)
Latch bias resistors	2 × 2 MΩ	2 × 2.2 MΩ (R5, R6)	E12 value in stock
Crystal loading	one 40 pF capacitor	C4 = C5 = 22 pF (both crystal terminals)	standard Pierce loading
Decoupling	5 × 100 nF in total	12 × 100 nF (C9, C12–C22) + 100 µF (C8)	local decoupling of the IC supply pins
Mechanical	—	4 × M3 holes, IC sockets for all 13 ICs, printed mode table	serviceability

3 System architecture

3.1 Measurement principle

Let f_{rev} be the revolution frequency and N_m the number of reflective marks on the rotor. The counters are cleared at the start of a window of length T_g = 1.000 s and count the clock presented on /CNT_CLK. The displayed number is therefore

$$N_{RPM} = 60 \cdot N_m \cdot f_{rev} \cdot T_g \qquad N_{RPS} = N_m \cdot f_{rev} \cdot T_g \qquad N_{CNT} = \Sigma \text{ pulses since reset} \tag{1}$$

With N_m = 1 and T_g = 1 s, N_{RPM} equals the speed in revolutions per minute. The count is an integer, so the displayed value moves in steps of 60 RPM (one count) in RPM mode and 1 RPS in RPS mode (Section 5.3).

3.2 Block diagram

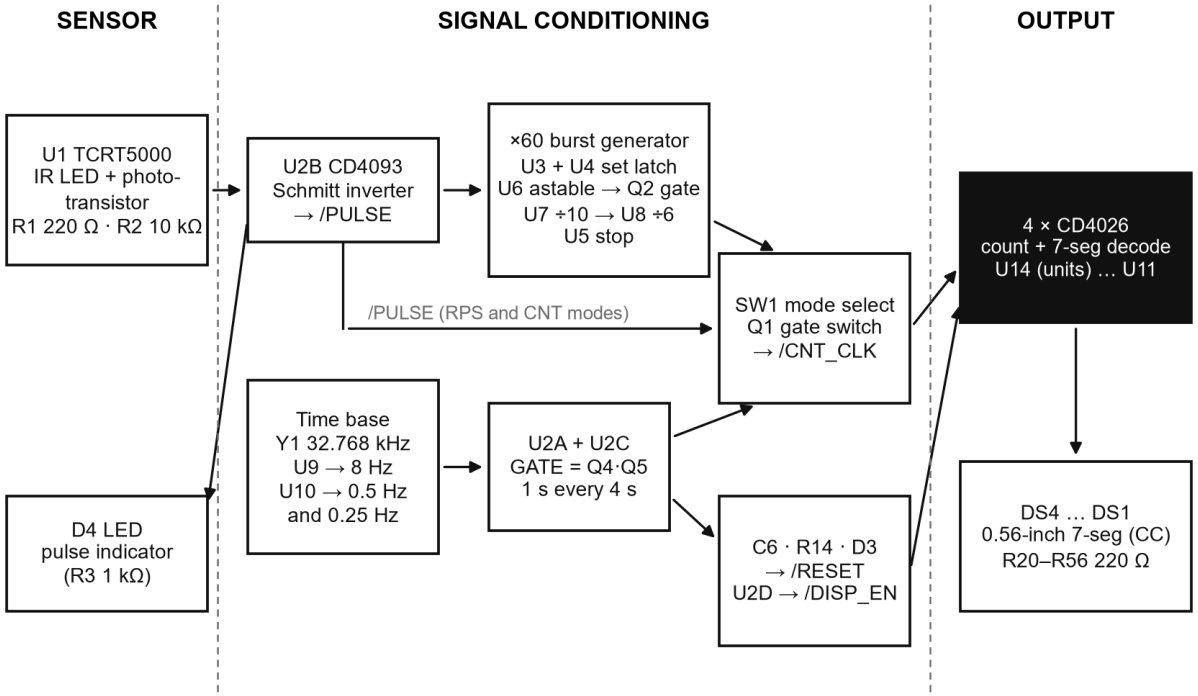


Figure 1. Functional block diagram grouped as sensor, signal conditioning and output. Reference designators as on the schematic.

3.3 Operating modes

The 4-way DIP switch SW1 selects the mode. Each switch n connects pin n to pin 9−n; the nets on each switch were read from the netlist [F]:

Table 3. SW1 switch functions [F].

Switch	Pins	Connects	Function when ON
1	1–8	/CNT_CLK ↔ /PULSE	raw pulses straight to the units counter, not gated (Counter mode)
2	2–7	/GATE ↔ /GATE_SEL	enables the 1 s window: gate transistor Q1, reset pulse (C6) and display blanking
3	3–6	/GATED_IN ↔ /BURST	gated input is the ×60 burst (RPM mode)
4	4–5	/GATED_IN ↔ /PULSE	gated input is the raw pulse (RPS mode)

Table 4. Operating modes (1 = ON). The same table is printed on the PCB silkscreen [F].

Mode	SW1 1-2-3-4	Display	Update	Resolution (1 mark)
RPM	0-1-1-0	revolutions per minute	every 4 s, blank 1 s while counting	60 RPM
RPS	0-1-0-1	revolutions per second	every 4 s, blank 1 s while counting	1 RPS
CNT	1-0-0-0	running total of marks	continuous, display always on	1 count

Forbidden setting. Switches 3 and 4 together connect /BURST (driven low by Q2 during a burst, pulled up by R17) to /PULSE (a CMOS totem-pole output), so the two outputs fight. Only the three settings of Table 4 are valid.

3.4 Key characteristics

Table 5. Summary of characteristics.

Characteristic	Value	Src
Supply	6.0 V nominal from 4 × AA; input marked 5–6 V	F
Sensor	Vishay TCRT5000 reflective optical sensor, IR LED at $I_F \approx 22$ mA	D, C
Counting window	1.000 s, once every 4.000 s; derived from 32.768 kHz by binary division only	C
Time-base error	< 0.005 % (crystal tolerance and load pulling, ±50 ppm assumed)	A, C
RPM resolution	60 RPM per count with one mark; $60/N_m$ RPM with N_m marks (display reads $N_m \times$ RPM)	C
RPM ceiling	4,809 RPM with C3 = 100 nF (as built); 21,859 RPM with C3 = 22 nF	C, S
Above the ceiling	revolutions are dropped; the reading settles at ≈ ½ of the true speed without warning	S
Display	4 digits, 0.56-inch red common-cathode 7-segment, 0000–9999	F
Update interval	4 s in RPM and RPS modes; continuous in counter mode	C
Working distance	datasheet: 0.2–15 mm (> 20 % of peak response), peak 2.5 mm; observed: works at 30 mm on a case fan with reflective tape	D, O
PCB	100.05 × 100.05 × 1.6 mm, 2 layers, 99 footprints, all through-hole	F
Parts	95 components: 13 ICs, 45 resistors, 22 capacitors, 4 displays	F
Parts cost	≈ 1 573.5 BDT including a PCB at ≈ 800 BDT (Section 9.3)	F, A

4 Circuit design

This section follows the signal through the circuit. Pin numbers and connections are taken from the netlist exported on 27 Sep 2026 [F]; the complete pin assignment of every IC is in Appendix D. The full schematic is Figure 2, with enlarged regions in Figures 3–5.

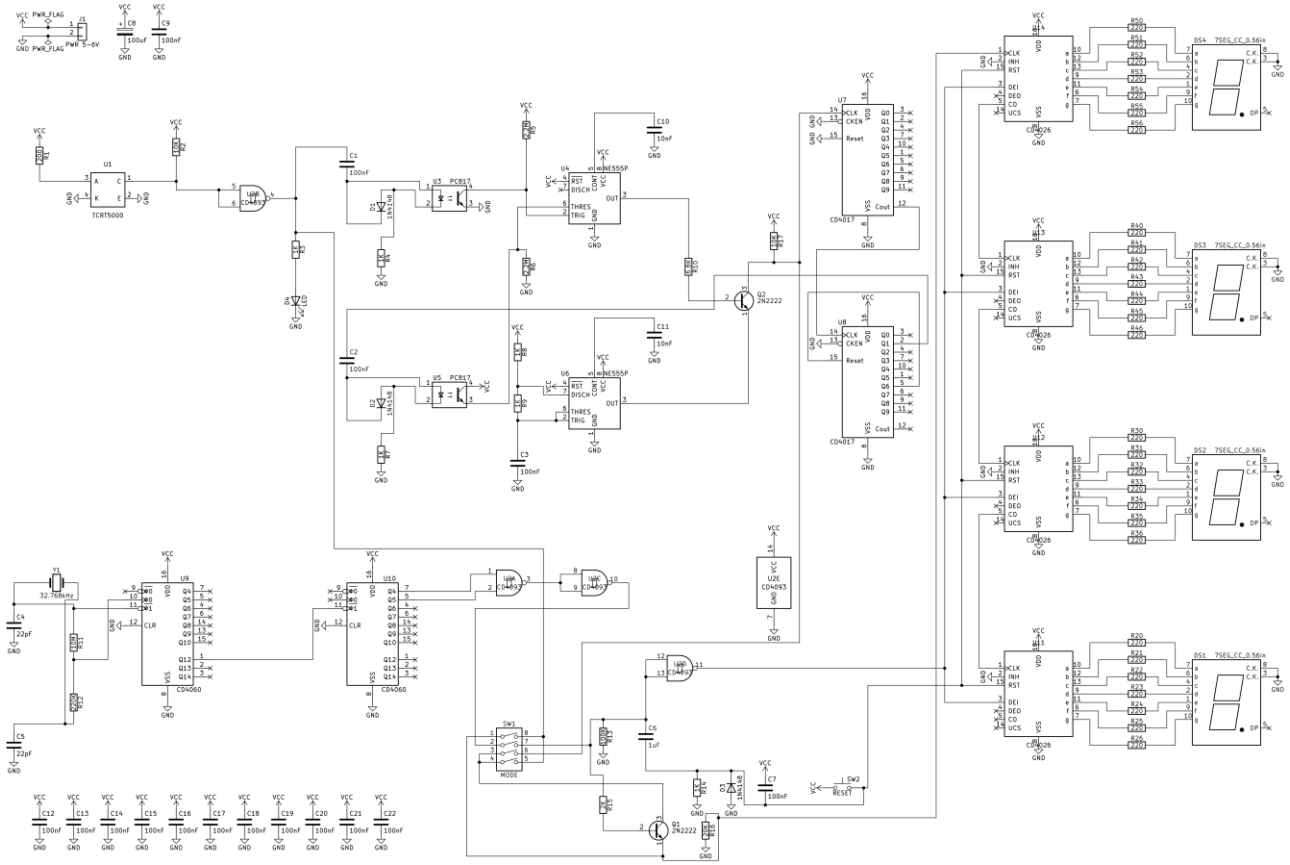


Figure 2. Complete schematic of tach_v1.0 (A3 sheet, KiCad 10, all connections drawn as wires, 0 net labels).

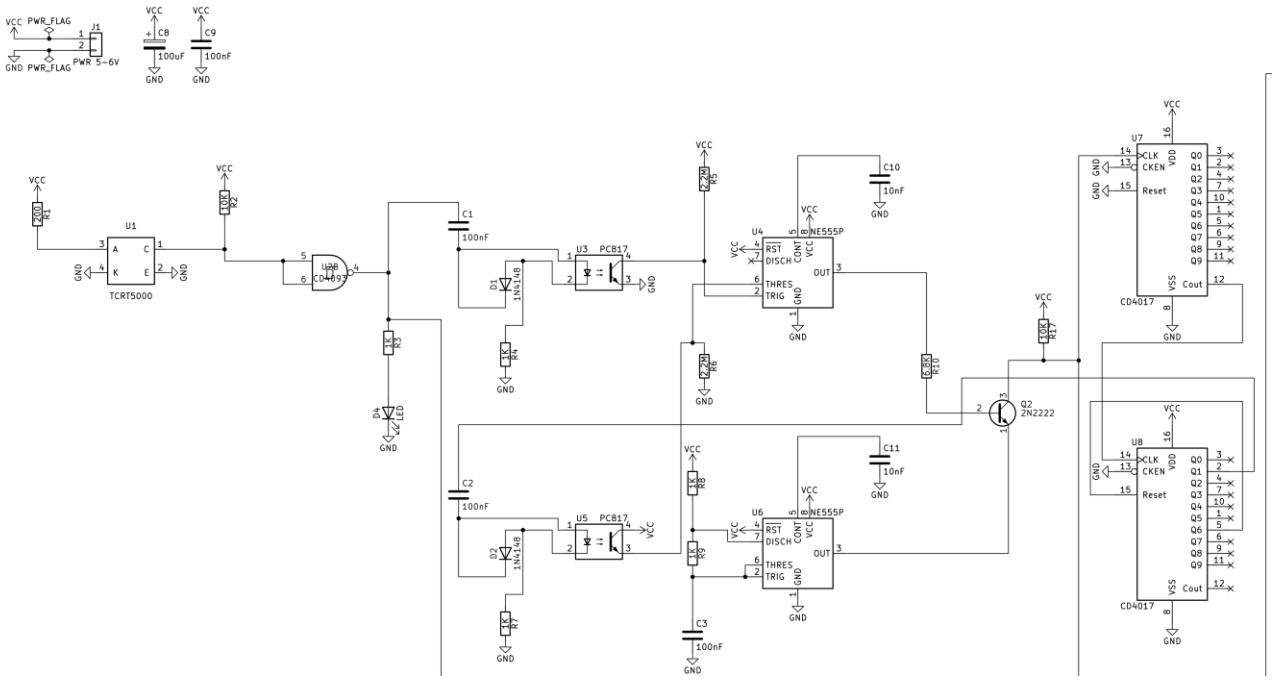


Figure 3. Schematic region A: power input, sensor and Schmitt trigger, set/stop optocouplers, latch U4, burst oscillator U6, burst gate Q2 and the ÷10/÷6 counters U7, U8.

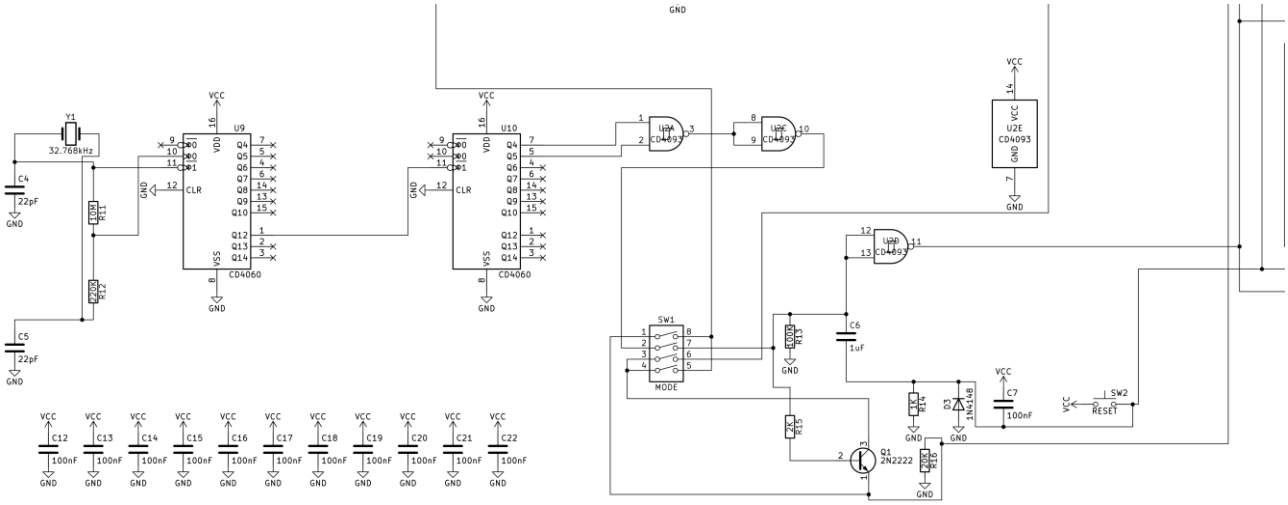


Figure 4. Schematic region B: crystal oscillator and dividers U9, U10, window logic U2A/U2C/U2D, mode switch SW1, reset network, gate transistor Q1 and the decoupling capacitors.

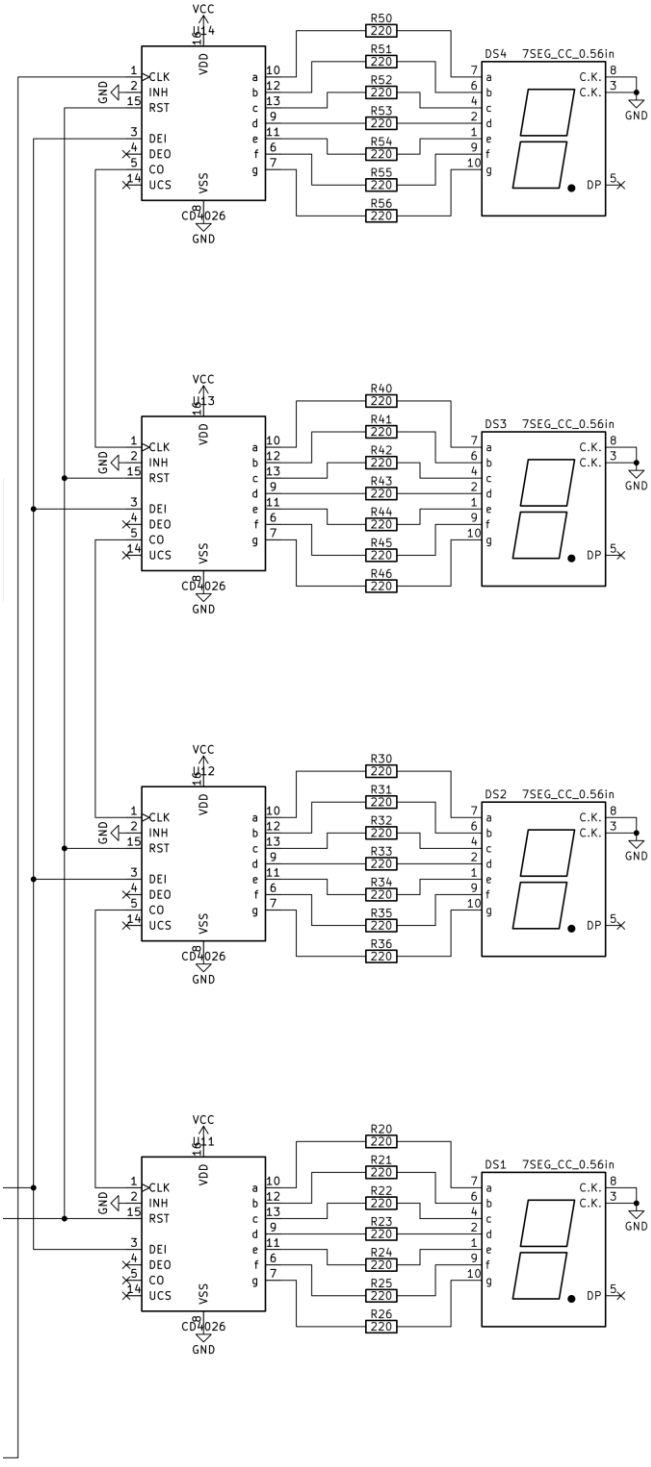


Figure 5. Schematic region C: counter chain U14 (units) → U13 → U12 → U11 (thousands), segment resistors and displays DS4…DS1.

4.1 Power supply and decoupling

The supply enters on J1 (2-pin 2.54 mm header footprint; on the built boards the battery leads are soldered directly into the J1 holes and no header is fitted). C8 (100 µF electrolytic) is the bulk capacitor; C9 and C12–C22 (12 × 100 nF) decouple the IC supply pins. The /VCC net has 35 pins and /GND 58 pins [F]. There is no reverse-polarity protection and no regulator: the CD4000-series ICs are specified for 3–18 V and the NE555 for 4.5–16 V [D], so the 4 × AA voltage (about 6.0–6.4 V fresh [O]) is used directly. Running at 6 V rather than 5 V increases the CD4026 segment drive and therefore display brightness (Section 4.8).

4.2 Optical sensor and pulse shaping

Sensor. U1 is a Vishay TCRT5000: an infrared emitter (950 nm) and a silicon NPN phototransistor side by side in a leaded package with a daylight-blocking filter; light reaches the detector only by reflection [D]. R1 feeds the IR LED from V_{CC}; R2 (10 kΩ) is the collector pull-up; the emitter is grounded.

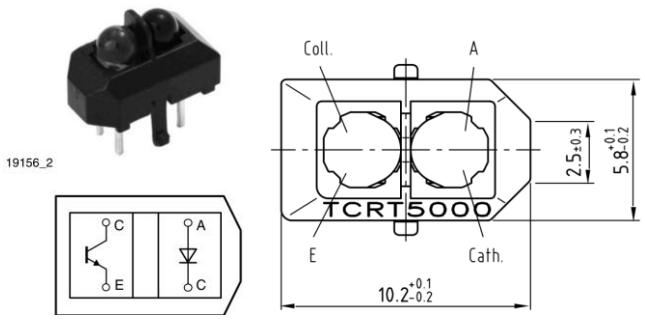


Figure 6. TCRT5000 package, internal symbol and top-view pin arrangement (Vishay datasheet, doc. 83760). Collector and anode are on the top edge, emitter and cathode on the bottom edge.

$$I_F = (V_{CC} - V_F) / R1 = (6.0 - 1.2) / 220\ \Omega \approx 21.8\ \text{mA} \quad \text{(range 21.6–22.0 mA for } V_F = 1.25\text{–}1.15\ \text{V)} \qquad (2)$$

$V_F \approx 1.2$ V at 20 mA is read from datasheet Fig. 4; the absolute maximum forward current is 60 mA [D]. The schematic value of R1 is 200 Ω (24.0 mA); 220 Ω is fitted (Section 9.2).

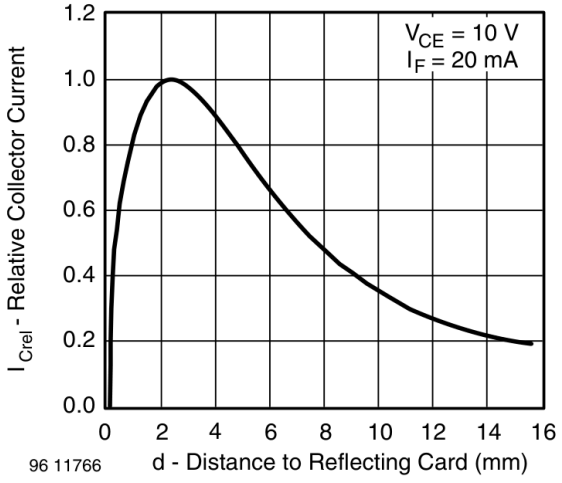


Fig. 9 - Relative Collector Current vs. Distance

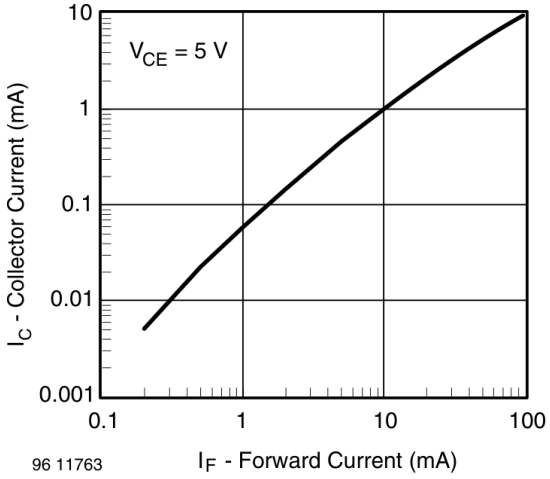


Fig. 6 - Collector Current vs. Forward Current

Figure 7. Datasheet Fig. 9, relative collector current versus distance (peak at 2.5 mm, V_{CE} = 10 V, I_F = 20 mA), and Fig. 6, collector current versus forward current (V_{CE} = 5 V) [D].

Schmitt trigger. /SENS_RAW drives both inputs (pins 5, 6) of gate U2B of the CD4093, which therefore acts as a Schmitt-trigger inverter with output /PULSE (pin 4). Over a dark surface the phototransistor is off, /SENS_RAW ≈ V_{CC} and /PULSE is low; when the reflective mark passes, the collector current pulls /SENS_RAW below the negative-going threshold V_N and /PULSE goes high. The CD4093 datasheet gives thresholds at 5 V and 10 V only; Table 6 interpolates them linearly to 6 V (both inputs tied, 25 °C).

Table 6. CD4093B trigger thresholds (inputs tied, T_A = 25 °C).

Quantity	5 V min / typ / max [D]	10 V min / typ / max [D]	6 V interpolated min / typ / max [C]
Positive-going threshold V_P	2.6 / 3.3 / 4.0 V	5.6 / 7.0 / 8.2 V	3.20 / 4.04 / 4.84 V
Negative-going threshold V_N	1.4 / 2.3 / 3.2 V	3.4 / 5.1 / 6.6 V	1.80 / 2.86 / 3.88 V
Hysteresis V_H (typ)	0.9 V	2.3 V	1.18 V

$$I_{C,on} = (V_{CC} - V_N) / R2 \qquad I_{C,off} = (V_{CC} - V_P) / R2 \qquad (3)$$

With R2 = 10 kΩ the mark is detected when I_C exceeds 0.314 mA (typical; 0.212–0.420 mA over the threshold spread) and released when it falls below 0.196 mA (0.116–0.280 mA) [C]. Full saturation of the output ($V_{CE(sat)} \leq$ 0.4 V [D]) needs $I_C \geq 0.56$ mA. The datasheet's typical collector current is 1 mA at I_F = 10 mA and 12 mm from its test surface [D]; the LED is driven at about twice that current here. Figure 8 shows the resulting transfer characteristic.

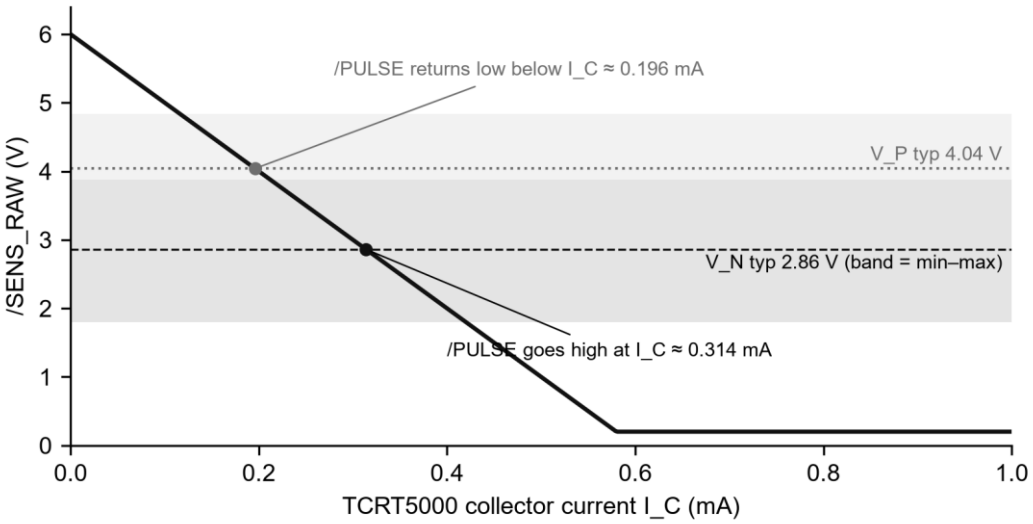


Figure 8. Sensor output /SENS_RAW versus phototransistor current with the CD4093 thresholds interpolated to 6 V (dashed: typical V_N; dotted: typical V_P; shaded: min–max spread).

Pulse indicator. D4 (5 mm LED) with R3 (1 kΩ) hangs on /PULSE and lights whenever the mark is in view, which makes aiming easy. Because a CMOS output is current-limited, the LED current is set by the gate rather than by R3 alone: solving the datasheet source characteristic of the CD4093 at 5 V against the LED load (V_F = 2.0 V assumed) gives ≈ 1.8 mA typical and ≈ 1.2 mA for a minimum-drive device [C, Appendix G].

4.3 Crystal time base

Y1 (32.768 kHz tuning-fork crystal) oscillates with the inverter inside U9 (CD4060) in a Pierce configuration: R11 (10 MΩ) biases the inverter from its output Φ_O (pin 10, /XTAL_DRV) to its input Φ_I (pin 11, /XTAL_IN); R12 (220 kΩ) isolates the output from the crystal (/XTAL_OUT); C4 and C5 (22 pF each) load the two crystal terminals. The series combination of C4 and C5 is 11 pF, to which PCB and pin capacitance add a few picofarads [C, A]. Pin 9 (inverted Φ_O) is unused. Both CD4060 reset inputs (pin 12) are tied to ground.

$$f_{U9\,Q12} = 32\,768\ \text{Hz} / 2^{12} = 8\ \text{Hz} \qquad f_{U10\,Q4} = 8\ \text{Hz} / 2^{4} = 0.5\ \text{Hz} \qquad f_{U10\,Q5} = 8\ \text{Hz} / 2^{5} = 0.25\ \text{Hz} \tag{4}$$

U9 Q12 (pin 1, /CLK_8HZ) clocks U10 through its oscillator input (pin 11). U10 Q4 (pin 7, /T_Q4_0.5HZ) and Q5 (pin 5, /T_Q5_0.25HZ) are the only divider outputs used; all other Q outputs of U9 and U10 are unconnected (Appendix C). No resistor or capacitor sets any time in the gate path, so the window length inherits the crystal accuracy directly.

4.4 Counting window and display enable

- **U2A** (pins 1, 2 → 3) = NAND(Q4, Q5) → /GATE_N.
- **U2C** (pins 8, 9 → 10) = NAND(/GATE_N, /GATE_N) = Q4 · Q5 → /GATE. Q4 is high for 1 s of every 2 s and Q5 for 2 s of every 4 s; both are high together for exactly 1 s of every 4 s.
- **SW1-2** connects /GATE to /GATE_SEL. With the switch off, R13 (100 kΩ) holds /GATE_SEL low.
- **U2D** (pins 12, 13 → 11) = NOT(/GATE_SEL) → /DISP_EN, wired to pin 3 (DISPLAY ENABLE IN) of all four CD4026. The display is dark while the window is open and lit otherwise. In counter mode /GATE_SEL is held low, so the display is permanently enabled.

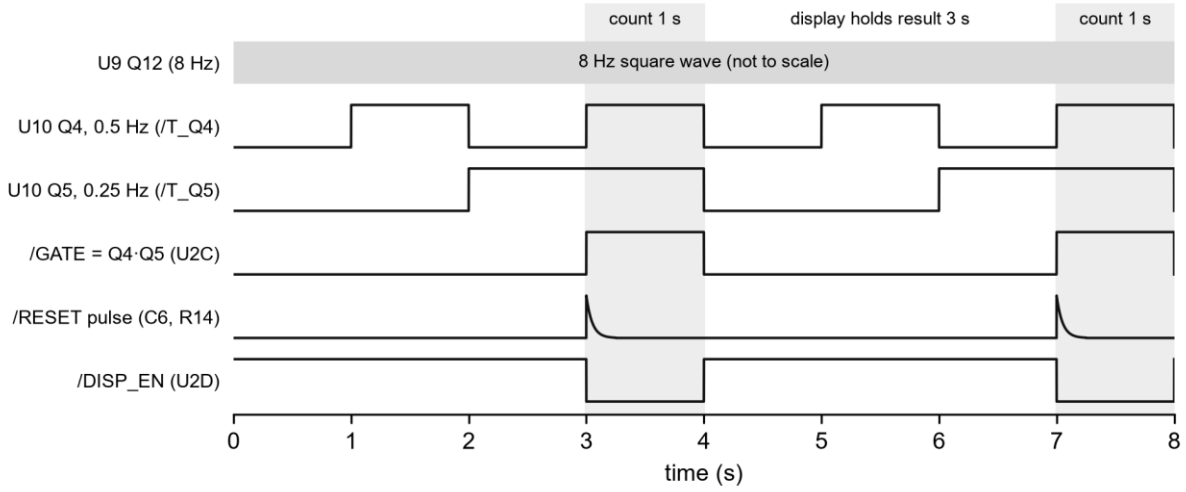


Figure 9. Window timing derived from the divider outputs. The reset pulse occurs on the rising edge of /GATE_SEL (Section 4.5).

4.5 Counter reset network

C6 (1 µF) couples /GATE_SEL to /RESET, which drives pin 15 (RESET, active high) of all four CD4026. R14 (1 kΩ) returns /RESET to ground, so the rising edge of the window produces a decaying reset pulse with $\tau = R14 \cdot C6 = 1.0$ ms [C]; the falling edge produces a negative spike that D3 (1N4148, cathode on /RESET) clamps to about −0.6 V. C7 (100 nF) from V_{CC} to /RESET gives a power-on reset pulse ($\tau = R14 \cdot C7 = 100$ µs), and SW2 connects /RESET to V_{CC} for a manual reset. The CD4026 needs a reset pulse of only 200 ns at 5 V [D].

Drive margin (finding). At the instant of the window edge C6 is uncharged, so the CD4093 output that drives /GATE_SEL (U2C, through SW1-2) sees R14 = 1 kΩ as its load. A CMOS output is current-limited: at $V_{DD} = 5$ V the CD4093 sources typically 3.2 mA at 2.5 V and 1.0 mA at 4.6 V (minimum 1.6 mA and 0.51 mA) [D]. Solving this characteristic against the 1 kΩ load gives a reset-pulse peak of ≈ 2.84 V for a typical device and ≈ 1.91 V for a minimum device at 5 V [C]. The CD4026 guarantees a logic high only above 3.5 V at 5 V (70 % of V_{DD}) [D]. The reset therefore relies on the typical input threshold (about half the supply) rather than on a guaranteed level. The built boards reset correctly [O], but the margin is not guaranteed. Scaling the network to C6 = 100 nF and R14 = 10 kΩ keeps $\tau = 1$ ms, cuts the load ten-fold and raises the calculated peak to ≈ 5.0 V (typical) and ≈ 4.7 V (minimum), above the guaranteed V_{IH} (Section 14).

C6 was fitted as a 1 µF electrolytic rather than the ceramic in the BOM. This is acceptable with its positive lead on /GATE_SEL (PCB pad 1): the only reverse voltage it sees is the −0.6 V allowed by D3 [C].

4.6 ×60 burst multiplier (RPM mode)

The multiplier turns each rising edge of /PULSE into exactly 60 rising edges on /BURST. It consists of a set path, a latch, a free-running oscillator, a gate transistor, a ÷60 counter and a stop path (Figure 10).

- **Set path.** C1 (100 nF) differentiates /PULSE into the LED of optocoupler U3 (PC817) through R4 (1 kΩ); D1 across the LED (anti-parallel) clamps the negative edge. $\tau = C1 \cdot R4 = 100$ µs [C]. The U3 transistor (collector pin 4) pulls /TRIG (U4 pin 2) low against R5 (2.2 MΩ to V_{CC}).
- **Latch.** U4 (NE555) is used as a set/reset flip-flop: TRIG (pin 2) below $V_{CC}/3$ sets the output /LATCH_Q (pin 3) high; THRESHOLD (pin 6) above $2V_{CC}/3$ resets it [D]. R6 (2.2 MΩ) holds THRESHOLD low; RESET (pin 4) is tied to V_{CC}; CONTROL (pin 5) is decoupled by C10 (10 nF); DISCHARGE (pin 7) is unused. PC817 CTR is at least 50 % at $I_F = 5$ mA [D]; the set pulse gives roughly 1.5–2.2 mA of LED current at 5 V [C], so the transistor can sink far more than the 2.7 µA flowing in R5.
- **Oscillator.** U6 (NE555) is a free-running astable with R8 = R9 = 1 kΩ and C3 = 100 nF (RESET tied to V_{CC}, CONTROL decoupled by C11 = 10 nF). $t_H = \ln 2 \cdot (R8 + R9) \cdot C3 = 138.6$ µs, $t_L = \ln 2 \cdot R9 \cdot C3 = 69.3$ µs, period 207.9 µs, f = 4,809 Hz (4,800 Hz with the common $1.44/((R8 + 2R9)C)$ approximation), duty cycle 66.7 % [C].
- **Gate transistor Q2** (2N2222). Base from /LATCH_Q through R10 (5.6 kΩ fitted, 6.8 kΩ drawn), emitter on the oscillator output /OSC, collector on /BURST with pull-up R17 (10 kΩ). While the latch is set and /OSC is

low, Q2 saturates and pulls /BURST low (base current ≈ 0.66 mA for a collector current of ≈ 0.58 mA, assuming the NE555 high level is ≈ 4.5 V at 6 V [C, A]). When /OSC is high, or the latch is reset, Q2 is off and R17 holds /BURST high. /BURST therefore copies /OSC during a burst and stays high otherwise.

- **÷60 counter.** U7 (CD4017) counts rising edges of /BURST (CLOCK pin 14; CLOCK INHIBIT pin 13 and RESET pin 15 grounded). Its CARRY OUT (pin 12, /CO10) is high for counts 0–4 and rises on the 9 → 0 transition, i.e. once every 10 clocks [D]. U8 (CD4017) counts /CO10 and has its Q6 output (pin 5) tied to its own RESET (pin 15), so it cycles through Q0…Q5 (÷6).
- **Stop path.** U8 Q1 (pin 2, /BURST_DONE) is differentiated by C2 (100 nF) into the LED of U5 (PC817) with R7 (1 kΩ) and clamp D2 (τ = 100 µs). The U5 transistor (collector at V_{CC}, emitter on /THRES) pulls U4 THRESHOLD above $2V_{CC}/3$ and resets the latch.

Why exactly 60. A burst ends on the rising edge of U8 Q1, so U8 rests at Q1 between bursts and U7 rests at 0 (its carry has just risen). The next burst must advance U8 through Q2, Q3, Q4, Q5, Q6 (which immediately resets it to Q0) and Q1: six carries of U7, i.e. 6 × 10 = 60 rising edges of /BURST. The number is fixed by counting; the oscillator frequency only sets how long the burst takes. A step-by-step logic model of this chain (the script that produced Figure 10) confirms 60 edges per burst [S].

Duration and latency. The first rising edge of /BURST follows the latch within one oscillator period, so a burst lasts between 59 and 60 periods after the set edge: 12.27–12.48 ms with C3 = 100 nF [C]. PC817 rise and fall times are 4 µs and 3 µs typical (18 µs maximum) [D], far shorter than the 138.6 µs high phase of /OSC in which the latch must be reset after the 60th edge, so no extra edge leaks through.

Power-up state. U7 and U8 have no power-on reset (U7 RESET is grounded; U8 is reset only by its own Q6). The first burst after power-up can therefore contain between 10 and 60 edges. Only the first reading after switch-on can be affected; the build procedure notes this [F].

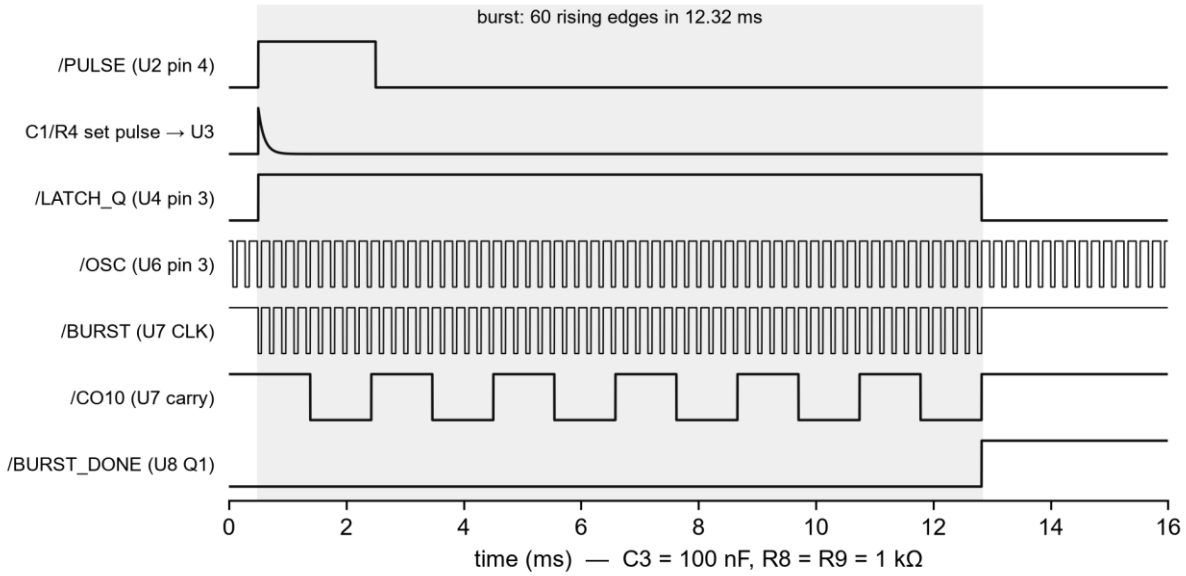


Figure 10. Derived timing of one burst (C3 = 100 nF). /BURST follows /OSC while /LATCH_Q is high; the 60th rising edge makes U8 Q1 rise, which resets the latch through C2 and U5.

4.7 Count gating transistor Q1

In RPM and RPS modes the counter clock /CNT_CLK comes from the emitter of Q1 (2N2222). The collector is the selected signal /GATED_IN (/BURST or /PULSE); the base is driven from /GATE_SEL through R15 (2.2 kΩ fitted, 2 kΩ drawn); R16 (22 kΩ fitted, 20 kΩ drawn) pulls the emitter to ground. When the window is closed the base is low, Q1 is off and R16 holds /CNT_CLK low. When the window is open Q1 passes the collector signal as an emitter follower: with V_{BE} = 0.7 V the high level is ≈ 5.3 V when the collector is the CMOS /PULSE and ≈ 5.0 V when it is /BURST through R17 (10 kΩ) [C]. Both exceed 70 % of 6 V = 4.2 V, and the CD4026 clock input has a Schmitt trigger [D]. The build procedure expects ≈ V_{bat} − 1 V on U14 pin 1 during the window, consistent with this [F].

In counter mode SW1-1 connects /PULSE directly to /CNT_CLK; Q1 is then off because /GATE_SEL is held low by R13.

4.8 Counting, decoding and display

U14 counts /CNT_CLK and drives DS4 (units, right-hand digit). Each CARRY OUT (pin 5) clocks the next counter: U14 → U13 (tens, DS3) → U12 (hundreds, DS2) → U11 (thousands, DS1). CLOCK INHIBIT (pin 2) of every CD4026 is grounded; DISPLAY ENABLE OUT (pin 4), UNGATED C (pin 14) and U11 CARRY OUT are unconnected. The display wraps from 9999 to 0000.

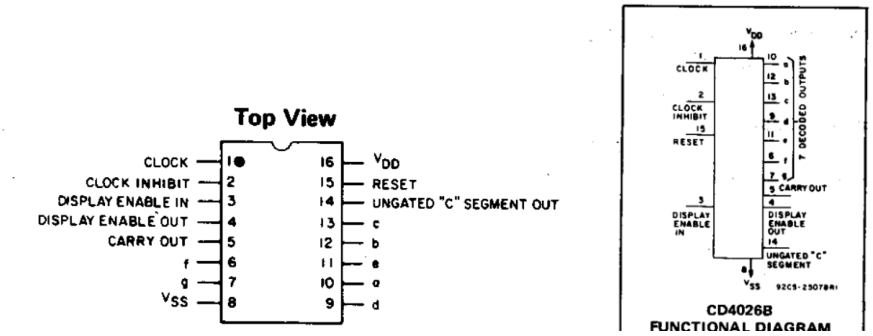


Figure 11. CD4026B terminal diagram and functional diagram (Texas Instruments SCHS031B) [D].

Table 7. Segment wiring. Counter U11 → DS1, U12 → DS2, U13 → DS3, U14 → DS4 [F].

Segment	CD4026 pin [D]	Display pin [F]	Resistor (DS1 / DS2 / DS3 / DS4)
a	10	7	R20 / R30 / R40 / R50
b	12	6	R21 / R31 / R41 / R51
c	13	4	R22 / R32 / R42 / R52
d	9	2	R23 / R33 / R43 / R53
e	11	1	R24 / R34 / R44 / R54
f	6	9	R25 / R35 / R45 / R55
g	7	10	R26 / R36 / R46 / R56
common cathode	—	3, 8 (to GND)	—
decimal point	—	5 (unconnected)	—

Segment current. The CD4026 segment outputs are specified as current sources: 3.2 mA typical (1.6 mA minimum) at V_O = 2.5 V and V_{DD} = 5 V [D]. With a red segment ($V_F \approx 2.0$ V [A]) in series with 220 Ω, the operating point is ≈ 3.0 mA typical and ≈ 1.7 mA minimum at 5 V [C]; the drive increases at 6 V. The chip, not the resistor, sets the current, which is why the value was reduced from the 470 Ω first drawn to 220 Ω: by the same method 470 Ω gives ≈ 2.5 mA typical, while 220 Ω lets the output deliver close to its full current [C]. Several web sources list the CD4026 segment pins in sequential order; the pin-out used here was checked against the TI datasheet [D].

4.9 Signal summary

Table 8. Principal signals.

Net	Nodes	Behaviour	Rate
/SENS_RAW	U1 C, R2, U2 5/6	≈ 6 V (dark) → ≤ 0.4 V (mark); switches at $I_C \approx 0.31$ mA	analogue
/PULSE	U2 pin 4	high while the mark is in view	1 per revolution
/CLK_8HZ	U9 pin 1	8 Hz square wave	8 Hz
/T_Q4_0.5HZ, /T_Q5_0.25HZ	U10 pins 7, 5	square waves	0.5 Hz, 0.25 Hz
/GATE, /GATE_SEL	U2 pin 10, SW1 pin 7	high 1 s of every 4 s	0.25 Hz
/DISP_EN	U2 pin 11	low during the window	0.25 Hz
/RESET	C6, R14, D3, CD4026 pin 15	pulse at window start, τ = 1 ms	0.25 Hz
/LATCH_Q	U4 pin 3	high for one burst	1 per revolution
/OSC	U6 pin 3	free-running, 4,809 Hz, 66.7 % duty	continuous
/BURST	Q2 C, U7 pin 14	60 pulses per revolution, otherwise high	4,809 Hz in bursts
/CO10, /BURST_DONE	U7 pin 12, U8 pin 2	÷10 carry; stop edge after 60 pulses	—
/CNT_CLK	Q1 E, U14 pin 1	gated clock, high ≈ 5.0–5.3 V	window only

5 Timing and performance analysis

5.1 Time base

The window is produced by binary division of the crystal frequency, so its relative error equals the crystal's. The fitted crystal's datasheet is not in the project (Section 13); for a typical 32.768 kHz tuning-fork crystal with ±20 ppm tolerance plus load-capacitance pulling of a similar order, a total below ±50 ppm is assumed [A]. This is 0.005 %, or 0.24 RPM at 4 800 RPM [C]: negligible beside quantisation.

5.2 Window and update rate

The window is open for 1.000 s and closed for 3.000 s, so a new reading appears every 4 s and the display is blank for 25 % of the time in RPM and RPS modes [C]. The display always shows the count of the most recent window.

5.3 Resolution and quantisation

A window of fixed length started at an arbitrary rotor phase captures either $\lfloor N_m\, f_{rev}\, T_g \rfloor$ or $\lceil N_m\, f_{rev}\, T_g \rceil$ marks. The reading therefore moves in steps of one count ($60/N_m$ RPM in RPM mode) and the error of any single reading is less than one count. If the true speed lies exactly half-way between two steps the error is ±½ count; just above or below a step it approaches one count. Table 9 lists the bound $60/(N_m \cdot \mathrm{RPM})$.

Table 9. Upper bound of the quantisation error (one count) relative to the true speed [C].

True speed (RPM)	1 mark (±60 RPM)	2 marks (±30 RPM)	4 marks (±15 RPM)
60	100.00 %	50.00 %	25.00 %
120	50.00 %	25.00 %	12.50 %
300	20.00 %	10.00 %	5.00 %
600	10.00 %	5.00 %	2.50 %
1,200	5.00 %	2.50 %	1.25 %
1,800	3.33 %	1.67 %	0.83 %
2,400	2.50 %	1.25 %	0.62 %
3,000	2.00 %	1.00 %	0.50 %
4,800	1.25 %	0.62 %	0.31 %

With N_m marks the display reads N_m × RPM and the RPM ceiling (Section 5.4) falls by the same factor. Figure 12 plots the bound.

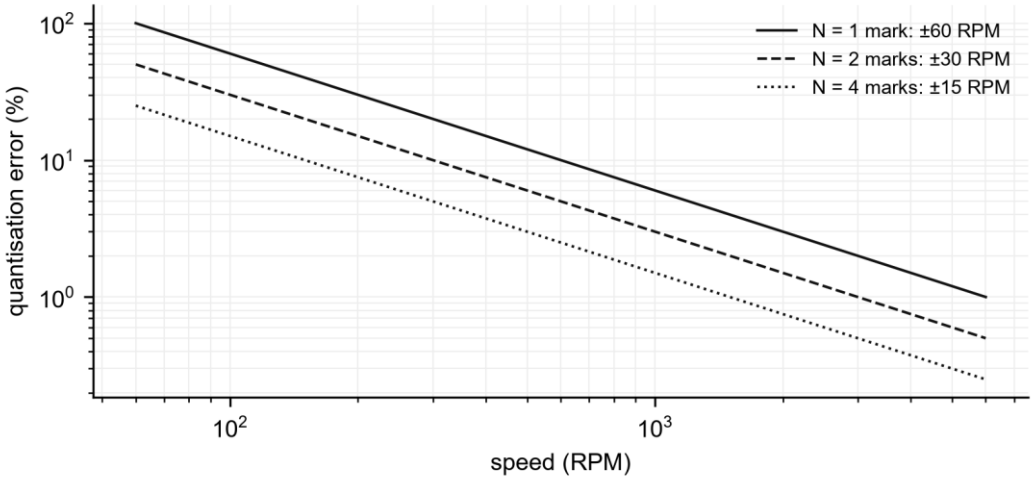


Figure 12. Quantisation error bound versus speed for one, two and four marks.

5.4 Maximum speed

A burst must end before the next revolution arrives: a set edge that arrives while the latch is still set does nothing, and that revolution is lost. With the burst lasting at most 60 oscillator periods the limit is

$$f_{rev,max} = f_{osc} / 60 \quad \Rightarrow \quad \mathrm{RPM}_{max} = 60 \cdot f_{rev,max} = f_{osc}\ \text{(numerically, in hertz)} \tag{5}$$

The RPM ceiling in RPM equals the oscillator frequency in hertz. With C3 = 100 nF this is 4,809 RPM (80.15 rev/s); allowing one extra period of latency gives 4,730 RPM. With C3 = 22 nF: 21,859 RPM [C]. Above the

ceiling every second revolution is lost and the display settles at about half the true speed (Section 5.5). The reading is steady and plausible, so the fault cannot be recognised from the display.

Table 10. RPM ceiling for C3 = 100 nF within ±20 % and at +80 % (the upper tolerance of Y5V/Z5U disc capacitors) [C]. R8 = R9 = 1 kΩ nominal.

C3	80 nF	90 nF	100 nF	110 nF	120 nF	180 nF
RPM ceiling	6,011	5,343	4,809	4,372	4,007	2,672

The dielectric of the fitted C3 is not recorded. With ±5 % resistors and ±20 % on C3 the ceiling lies between 3,817 and 6,328 RPM [C]. The ceiling of an individual board can be found by measuring the frequency at U6 pin 3: that frequency in hertz is the ceiling in RPM.

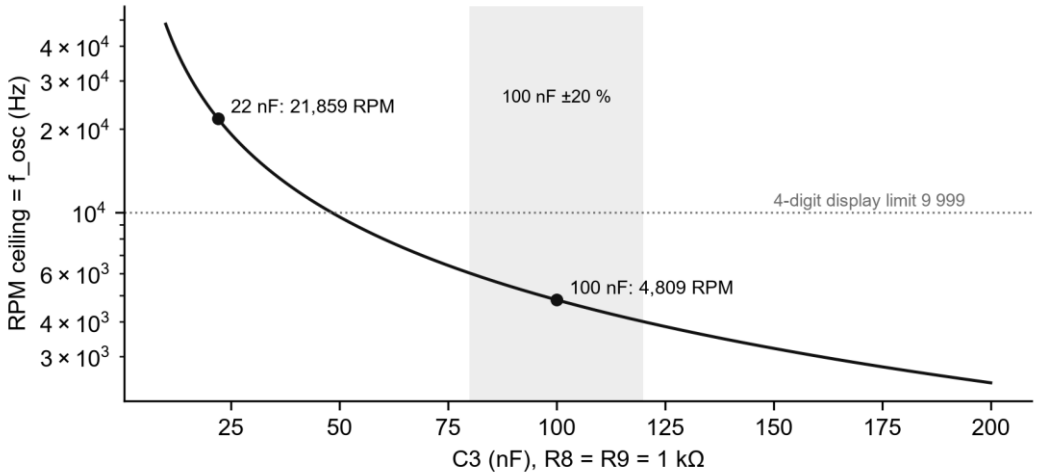


Figure 13. RPM ceiling as a function of C3.

5.5 Behavioural simulation

A SPICE simulation was not used for the counting chain: no reliable models of the CD4026, CD4017 and CD4060 were available, and the quantities that decide accuracy are logic timing, not analogue behaviour. Instead the chain was modelled event by event (analysis/sweep.py) [S]:

- window open 1.000 s once every 4 s; counters cleared at the start of each window;
- RPM mode: each revolution produces 60 counter clocks at intervals of $1/f_{osc}$; a revolution arriving while the previous burst is still running is dropped (latch already set);
- bursts that straddle a window edge contribute only the clocks inside the window;
- input frequency 1.0–100.0 Hz in 0.5 Hz steps; 12 rotor phases × 6 windows per frequency; the range of readings is reported.

f_{osc} was set to 4 800 Hz (the 1.44 approximation; the ln 2 value is 4 809 Hz) for C3 = 100 nF and 21 818 Hz for C3 = 22 nF. Not modelled: the optical front end, component tolerances, the reset-pulse duration and the power-up state of U7/U8.

Table 11. Simulated readings (minimum–maximum over 12 phases × 6 windows) [S]. Full data at 0.5 Hz steps: analysis/sweep.json.

Input (Hz)	True RPM	Reading, C3 = 100 nF	Revs lost (%)	Reading, C3 = 22 nF	Revs lost (%)
1	60	60–60	0.0	60–60	0.0
1.5	90	60–120	0.0	60–120	0.0
2	120	120–120	0.0	120–120	0.0
2.5	150	120–180	0.0	120–180	0.0
5	300	300–300	0.0	300–300	0.0
10	600	600–600	0.0	600–600	0.0
20	1,200	1,200–1,200	0.0	1,200–1,200	0.0
30	1,800	1,800–1,800	0.0	1,800–1,800	0.0
40	2,400	2,400–2,400	0.0	2,400–2,400	0.0
50	3,000	3,000–3,000	0.0	3,000–3,000	0.0
60	3,600	3,600–3,600	0.0	3,600–3,600	0.0
70	4,200	4,200–4,200	0.0	4,200–4,200	0.0
75	4,500	4,500–4,500	0.0	4,500–4,500	0.0
78	4,680	4,680–4,680	0.0	4,680–4,680	0.0
79	4,740	4,740–4,740	0.0	4,740–4,740	0.0
79.5	4,770	4,770–4,770	0.0	4,740–4,800	0.0

Input (Hz)	True RPM	Reading, C3 = 100 nF	Revs lost (%)	Reading, C3 = 22 nF	Revs lost (%)
80	4,800	3,840–4,800	6.0	4,800–4,800	0.0
80.5	4,830	2,400–2,426	50.0	4,800–4,860	0.0
81	4,860	2,401–2,456	50.0	4,860–4,860	0.0
82	4,920	2,460–2,460	50.0	4,920–4,920	0.0
85	5,100	2,524–2,576	50.0	5,100–5,100	0.0
90	5,400	2,700–2,700	50.0	5,400–5,400	0.0
95	5,700	2,830–2,870	50.0	5,700–5,700	0.0
100	6,000	3,000–3,000	50.0	6,000–6,000	0.0

Two effects are visible. First, at integer input frequencies every window contains a whole number of periods and the reading is exact; at half-integer frequencies it alternates between the two neighbouring multiples of 60 (for example 90 RPM reads 60 or 120). Integer-frequency rows must therefore not be read as evidence of accuracy. Second, with C3 = 100 nF, 6 % of revolutions are lost at 80.0 Hz and 50 % from 80.5 Hz upwards, where the reading drops to about half; with C3 = 22 nF no revolution is lost up to 100 Hz.

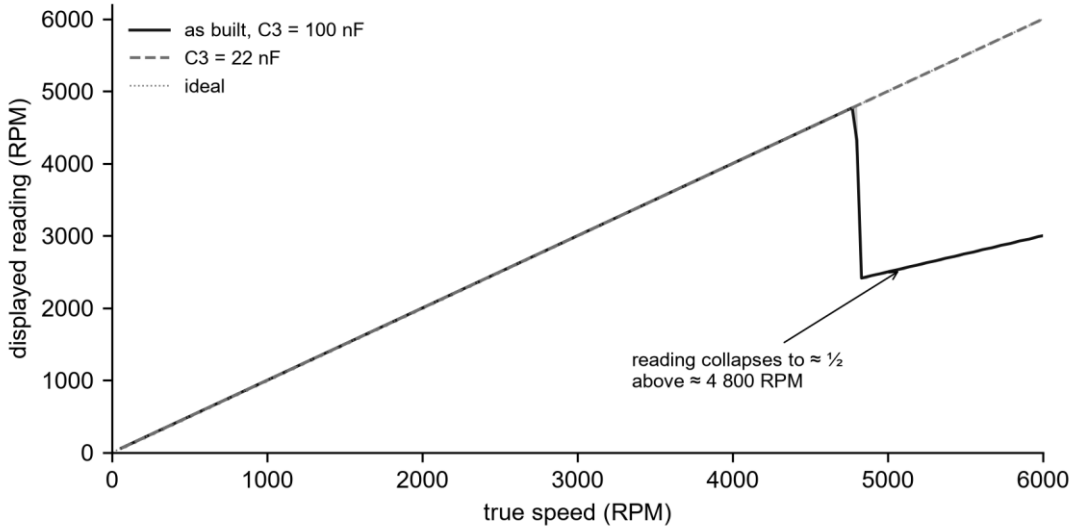


Figure 14. Simulated reading versus true speed; bands show the spread over rotor phase.

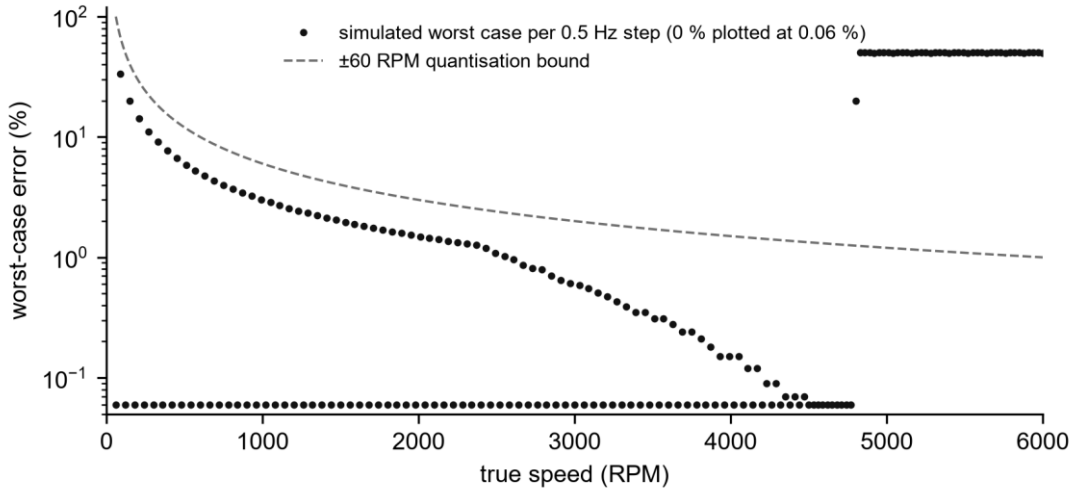


Figure 15. Simulated worst-case error per input frequency (C3 = 100 nF) against the one-count bound 60/RPM. Exact (0 %) results are plotted at 0.06 % so they remain visible on the log axis.

5.6 Second-order effects

- **Reset overlap.** The counters are held in reset for a fraction of the reset time constant (at most about 1 ms) at the start of the window. If a burst is running at that moment up to ≈ 5 of its clocks are not counted; this is below 0.1 revolution and is covered by the one-count quantisation [C].
- **First burst after power-up** may be 10–60 counts (Section 4.6).

- **Burst straddling the window edge.** A burst that starts less than 12.5 ms before the window closes contributes only part of its 60 counts; the complementary part of a burst that started just before the window opened is counted instead. This is part of the quantisation, not an extra error, and is included in the simulation [S].
- **Optical front end.** Missing pulses (too little reflected light) make the instrument under-read; extra pulses (ambient infrared, a second reflective feature, polished shafts) make it over-read. Neither is modelled; the CD4093 hysteresis rejects slow noise but there is no gain stage and no adjustable threshold.

5.7 Error budget

Table 12. Error budget.

Source	Magnitude	Type	Src
Time base (crystal)	< ±0.005 % (±50 ppm assumed)	systematic	A, C
Quantisation (1 mark)	< ±60 RPM (±1 count); ±30 RPM at half-way speeds	random with rotor phase	C, S
Reset overlap	≤ ≈ 5 burst clocks (< 0.1 revolution)	systematic, under-reads	C
Revolutions lost above the ceiling	50 % above ≈ 4,809 RPM (C3 = 100 nF)	gross, silent	S
First reading after power-up	10–60 counts per burst	one-off	C
Sensor misses / extra pulses	not quantified	gross	—

5.8 Power budget (estimate)

Table 13. Estimated supply current. Not measured.

Load	Current at 6 V	Basis	Src
TCRT5000 IR LED	≈ 22 mA continuous	Eq. (2)	C
NE555 U4 and U6, quiescent	2 × (2–3 mA typ, 5–6 mA max at 5 V)	datasheet supply current	D
U6 timing network (R8, R9)	≈ 3.0 mA average	1.5 mA during t_H, 6 mA during t_L	C
Displays	≈ 1.7–3.0 mA per lit segment (5 V); up to 28 segments ≈ 47–85 mA	Section 4.8	C
Pulse LED D4	≈ 1–2 mA while the mark is in view	Section 4.2	C
CMOS logic (CD40xx)	µA quiescent	datasheet I_{DD}	D
Total	≈ 30 mA (display blank) to ≈ 120 mA (8888 lit)	sum	C

6 Schematic capture

The schematic is a single A3 sheet (tach_v1.0.kicad_sch, KiCad 10 file format 20260306). It contains 190 placed symbols: the 95 components, 54 GND and 35 VCC power symbols and 2 PWR_FLAG symbols; 394 wires, 44 junctions and 53 no-connect flags. Every signal connection is a drawn wire: the sheet has no net labels [F].

Table 14. Symbols used [F].

Library symbol	Count	Library symbol	Count
Device:R	45	Timer:NE555P	2
Device:C	21	Transistor_BJT:PN2222A (2N2222 fitted)	2
Device:C_Polarized	1	4xxx:4017	2
tach_lib:CD4026 (custom)	4	4xxx:4060	2
Display_Character:LTS-6980HR (CC)	4	4xxx:HEF4093B (U2, 5 units incl. power)	1
Diode:1N4148	3	tach_lib:TCRT5000 (custom)	1
Isolator:PC817	2	Device:LED / Device:Crystal	1 / 1
Connector_Generic:Conn_01x02	1	Switch:SW_DIP_x04 / Switch:SW_Push	1 / 1

Custom library tach_lib.kicad_sym holds CD4026 and TCRT5000, neither of which is in the KiCad standard library. The TCRT5000 symbol has an error: its pins 3 and 4 are named A and K, but on the footprint (and on the part) pad 3 is the cathode and pad 4 the anode (Section 11). **Custom footprint library tach_fp.pretty** holds TCRT5000_NoClip, a copy of Vishay_TCRT5000 without its un-numbered clip holes, which broke the Specctra DSN export used for autorouting; it carries the Vishay_TCRT5000 3D model at zero offset.

Net names. The schematic was first drawn with net labels, and the PCB carries those names (/PULSE, /BURST, …). On 2026-09-23 it was redrawn with wires only. The exported netlist was checked to be pin-for-pin identical to the labelled version (95 parts, 149 nets), but without labels KiCad assigns automatic names (Net-(C1-Pad1),

…). The PCB was deliberately not updated, because it is the board that was ordered. The only consequence is the 199 net-name warnings of the parity check (Section 7.10).

ERC. kicad-cli sch erc with all severities: 0 errors, 0 warnings (27 Sep 2026) [F].

7 PCB design

7.1 Outline, stack-up and mounting

The outline (Edge.Cuts) is four lines and four 5 mm-radius arcs enclosing 100.05 × 100.05 mm (9,988.5 mm²) [F, C]. The stack-up in the Gerber job file is 2 copper layers of 35 µm on 1.51 mm FR4, 10 µm solder mask each side, 1.6 mm finished thickness [F]. All 99 footprints are on the top side and all are through-hole. Four unplated 3.2 mm holes accept M3 screws; their positions were chosen by searching 629 candidate points against pads, tracks, courtyards and silkscreen for the largest copper-free radius, because the corner positions collide with J1 (top-left) [F].

Table 15. Mounting holes (3.2 mm NPTH). Radii from the placement search [F].

Hole	X (mm)	Y (mm)	From top-left corner (mm)	Copper-free radius (mm)
H1	104.38	63.08	4.41, 13.10	4.36
H2	196.13	62.67	96.16, 12.70	3.76
H3	103.68	145.07	3.71, 95.09	3.70
H4	196.28	145.07	96.31, 95.09	3.60

7.2 Placement

The board is laid out in functional zones that follow the block diagram (Figures 16–18). Along the top edge: power input J1 with C8, RESET button SW2, mode switch SW1, sensor U1 and the pulse LED D4 (top-right). Middle band: time base U9/U10 with the crystal on the left, U2, the burst multiplier (U3–U6, Q1, Q2) in the centre, and U7/U8 on the right. Lower band: the four CD4026 in a row, then the 28 segment resistors in four groups of seven, then DS1…DS4 along the bottom edge. The digits read left to right as thousands to units.

Table 16. Positions of key parts (footprint origin; for DIP footprints the origin is pin 1). All positions: Appendix A [F].

Ref	Part	X (mm)	Y (mm)	Rot (°)
J1	PWR 5-6V	103.50	54.00	0
SW2	RESET	121.00	53.00	0
SW1	MODE	133.00	61.50	90
U1	TCRT5000	147.50	53.00	0
D4	LED	190.00	56.00	0
Y1	32.768kHz	116.00	78.50	0
U9	CD4060	103.50	69.00	0
U10	CD4060	125.00	69.00	0
U2	CD4093	136.00	69.00	0
U4	NE555P	169.00	68.00	0
U6	NE555P	169.00	80.00	0
U7	CD4017	168.00	61.50	90
U8	CD4017	190.50	68.00	0
U11	CD4026	104.50	110.00	90
U14	CD4026	176.50	110.00	90
DS1	7SEG_CC_0.56in	109.00	145.50	90
DS4	7SEG_CC_0.56in	181.00	145.50	90

7.3 Design rules

Table 17. Board design rules [F: tach_v1.0.kicad_pro, zone settings].

Rule	Value	Rule	Value
Net class Default: clearance	0.20 mm	Minimum track width	0.20 mm
Net class Default: track / via	0.20 mm / 0.60 mm, drill 0.30 mm	Minimum via diameter / drill	0.50 mm / 0.30 mm
Copper to board edge	0.50 mm	Minimum annular ring (vias)	0.10 mm
Hole clearance / hole to hole	0.25 mm / 0.25 mm	Minimum text height / thickness	0.80 mm / 0.08 mm
Zone clearance / minimum width	0.30 mm / 0.25 mm	Thermal relief gap / spoke	0.50 mm / 0.50 mm

One custom rule (tach_v1.0.kicad_dru) lowers the minimum number of resolved thermal spokes on /GND pads from 2 to 1. A few ground pads sit where their own ground tracks leave room for only one pour spoke; every ground pad is also connected by a routed track, so one spoke is sufficient, and thermal relief is kept for hand soldering [F]. (KiCad silently ignores a constraint written as "(min 1)"; the working form is "(constraint min_resolved_spokes 1)".)

7.4 Routing

Routing was done with the Freerouting autorouter through a Specctra DSN/SES round trip. The DSN export of the KiCad automation server (Konnect) rejected the board's rounded-rectangle and oval pads, so KiCad's own exporter was used. The pipeline (routing/*.py): export the DSN with KiCad's own exporter; add a Power class so /VCC and /GND are routed at 0.40 mm (KiCad's DSN export ignores project net classes; 0.50 mm did not complete); lock already-routed wiring for incremental passes; route; import the session into a copy of the board, add the ground pours, fill and check with DRC before touching the live board [F]. Four routing passes (r1–r4) were run.

Table 18. Track statistics [F].

Width	Segments	Length (mm)	Nets
0.25 mm	648	2,743.8	all signal nets
0.40 mm	294	1,050.6	/GND (657.6 mm), /VCC (393.0 mm)
Total	942	3,794.4	F.Cu 1 553.5 mm, B.Cu 2 240.9 mm

There are 14 vias, all 0.60 mm with 0.30 mm drill, all on signal nets (/LATCH_Q 2; /LED_IND, /SEG3_g, /SET_A, /BURST, /CARRY_TENS, /DISP_EN, /SEG2_e, /GATED_IN, /T_Q4_0.5HZ, /SENS_RAW, /SEG3_d, /SEG4_c 1 each). Longest signal nets: /RESET 155.1 mm, /DISP_EN 122.8 mm, /BURST 94.8 mm, /CNT_CLK 90.4 mm, /PULSE 83.5 mm [F].

7.5 Copper pours

Both copper layers carry a /GND zone (clearance 0.30 mm, minimum width 0.25 mm, thermal relief 0.50 mm gap and spoke). Filled areas: F.Cu 6 796.2 mm² (68.0 % of the board), B.Cu 5 952.9 mm² (59.6 %) [F, C].

7.6 Pads and drills

Table 19. Drill table: 388 plated holes (374 pad holes + 14 vias) and 4 unplated [F].

Drill (mm)	Plating	Holes	Used by
0.30	PTH (via)	14	vias
0.50	PTH	2	Y1 (crystal)
0.75	PTH	6	Q1, Q2 (TO-92 inline)
0.80	PTH	354	resistors, capacitors, diodes, ICs, displays, SW1
0.90	PTH	2	D4 (5 mm LED)
1.00	PTH	6	J1, U1 (TCRT5000)
1.10	PTH	4	SW2 (6 mm push button)
3.20	NPTH	4	H1–H4

The smallest annular ring on the board is 0.15 mm on the six Q1/Q2 pads (1.05 mm pad on a 0.75 mm drill), which equals JLCPCB's minimum; every other pad has at least 0.25 mm (314 pads have 0.40 mm) [F, C].

7.7 Silkscreen

Reference designators are hidden on the silkscreen and kept on the F.Fab layer. Instead, 78 free text items give component values: 13 at 0.8 mm, 49 at 1.0 mm and 16 at 1.4 mm height; nothing is below 0.8 mm height or 0.15 mm stroke [F]. The labels were placed by a constraint solver (tools/silkscreen/labels.py): each label is anchored on its footprint's silk-outline centroid, kept at least 0.25 mm from pads, 2.2 mm from mounting holes and 1.0 mm from the edge, and never more than 2.2 mm from its own part. Other features: a mode table (RPM 0110, RPS 0101, CNT 1000) drawn with 9 lines near U10; "220R" at each resistor group; a mirrored OptoTACH wordmark and maker mark on B.Silkscreen (57 shapes), imported as flattened SVG outlines. The silkscreen shows the drawn values, which differ from the fitted parts for R1, R10, R15 and R16 (Section 9.2).

7.8 3D model

All 95 component footprints carry a 3D model (the mounting holes need none). Figures 19–20 are rendered from the board file with kicad-cli.

7.9 Layer plots and renders

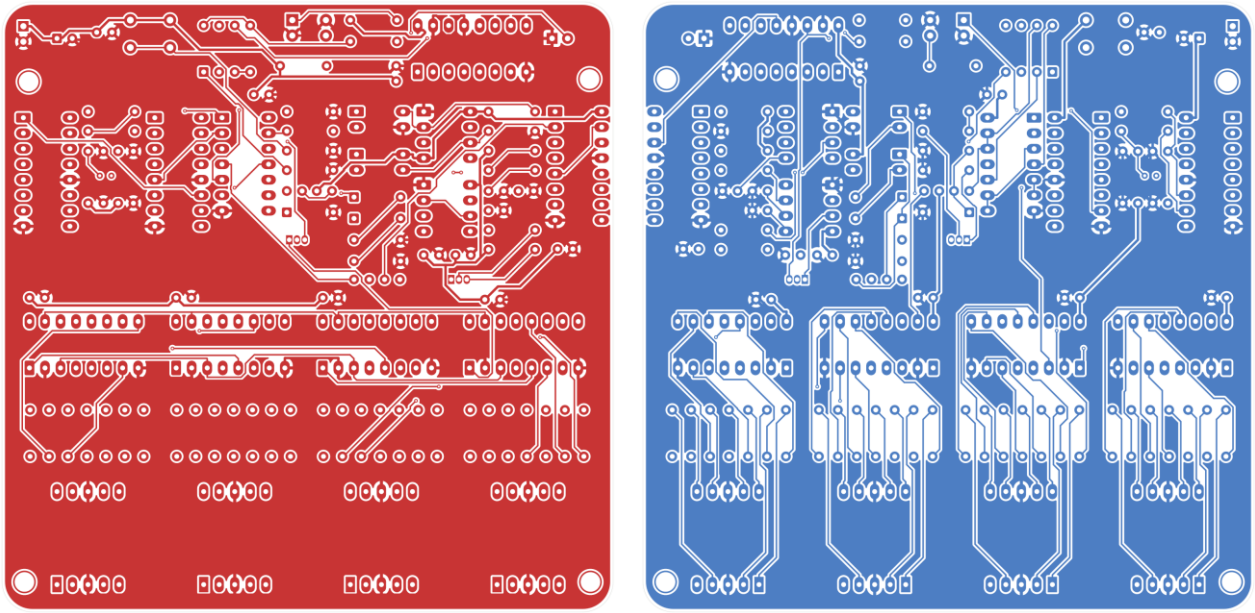


Figure 16. Top copper (left) and bottom copper viewed from the bottom (right), with /GND pours.

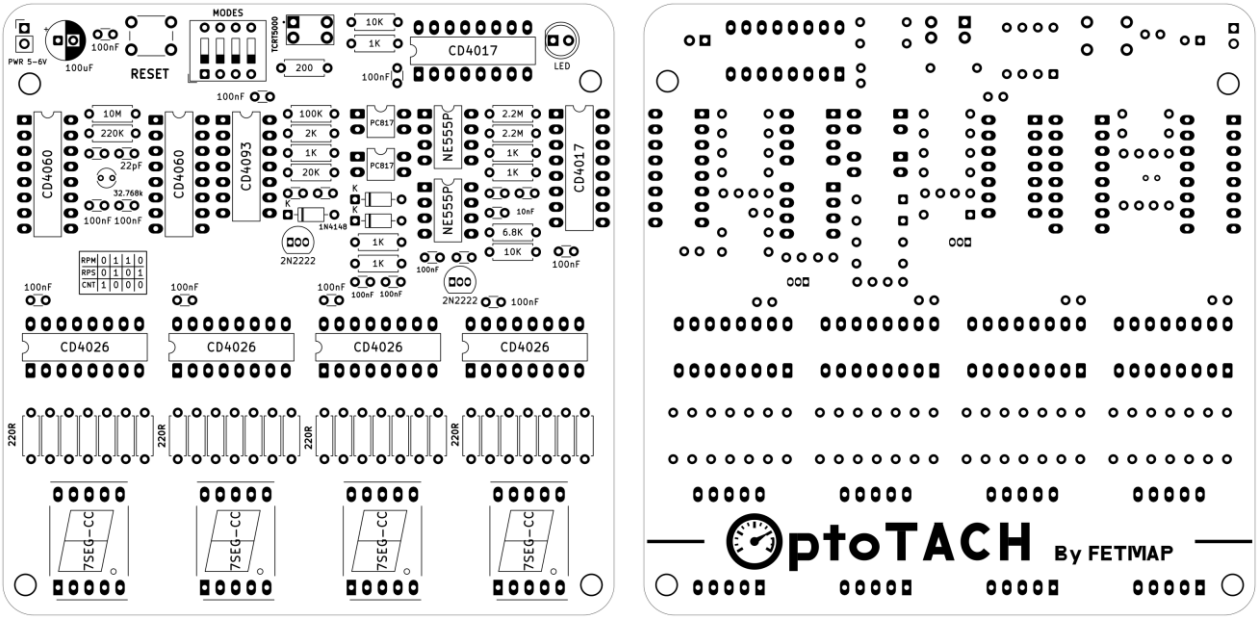


Figure 17. Top silkscreen with solder mask openings (left) and bottom silkscreen viewed from the bottom (right).

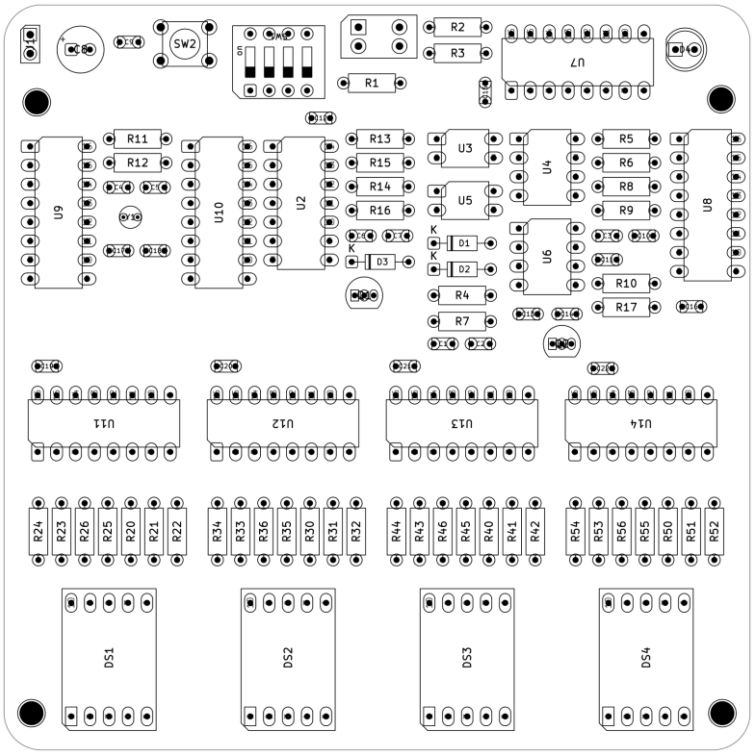


Figure 18. Fabrication layer (F.Fab) with reference designators and pad outlines.

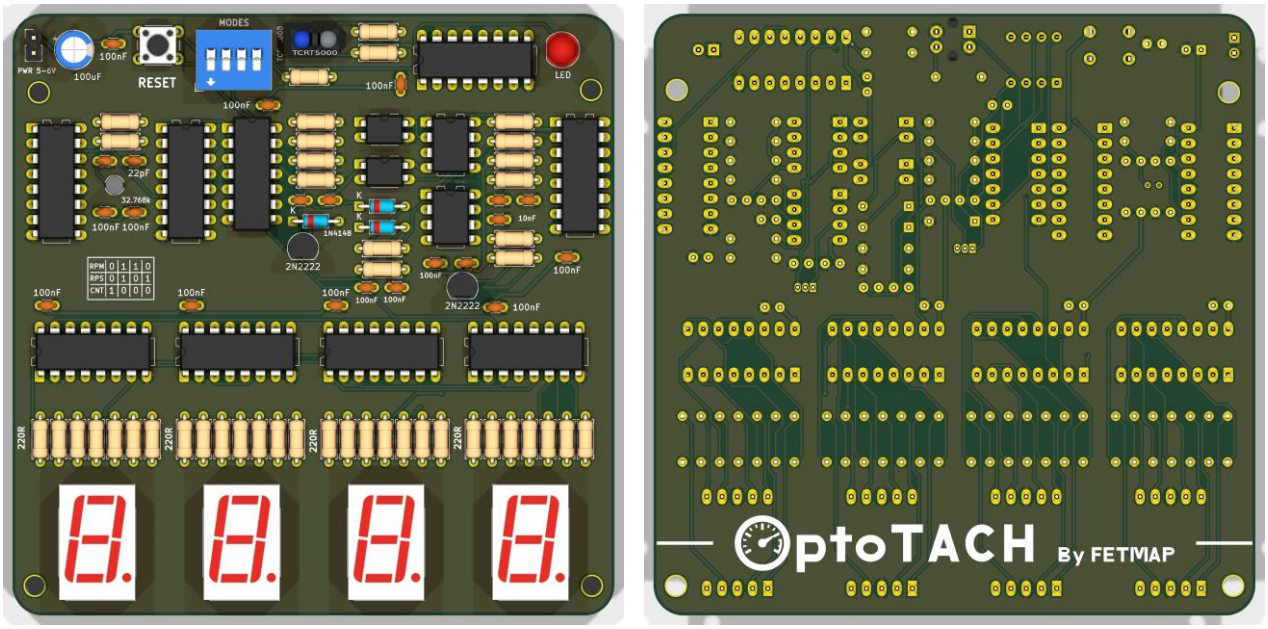


Figure 19. 3D render, top and bottom (kicad-cli pcb render).

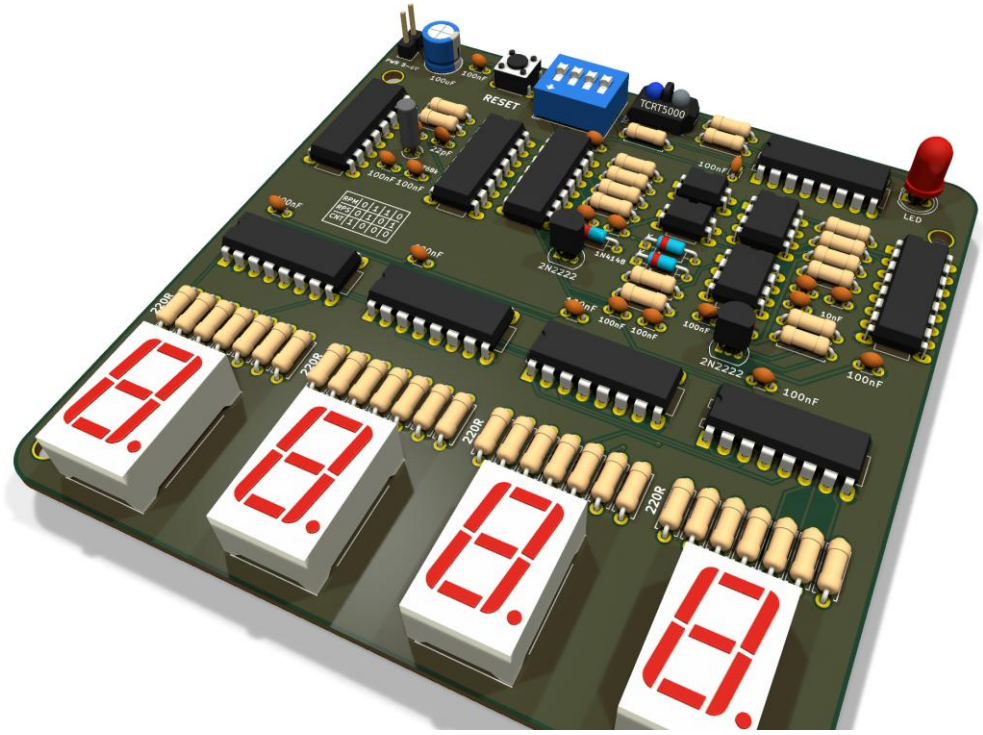


Figure 20. 3D render, perspective view.

7.10 Design rule check and connectivity verification

kicad-cli pcb drc was run on 27 Sep 2026 with all severities and the schematic-parity check enabled [F]:

Table 20. DRC summary [F].

Category	Count	Detail
Errors	0	—
Unconnected items	0	—
lib_footprint_mismatch (warning)	22	U2–U14 DIP footprints, DS1–DS4, Q1, Q2, SW1, SW2, D4: board copy differs from library copy after 3D models were added
lib_footprint_issues (warning)	4	H1–H4: MountingHole_3.2mm has no library copy in this installation
silk_overlap (warning)	4	the four “7SEG-CC” labels cross the display outlines
Parity: net_conflict	199	pad net names differ from the auto-named nets of the redrawn schematic (names only, see below)
Parity: footprint_symbol_field_mismatch	95	empty Description field on the PCB footprints
Parity: extra_footprint	4	H1–H4 exist only on the board

To show that the net-name warnings do not hide a wiring difference, the pad connectivity was read from the board with KiCad's Python API and compared with the schematic netlist as sets of (reference, pin) pairs. Both contain 149 nets and 372 connected pins, and the two partitions are identical: every PCB net has exactly one schematic net with the same members and vice versa (53 of them are single-pin no-connect nets whose names match; 96 are multi-pin). No footprint differs in value or footprint name from its symbol [F, C]. An independent check made earlier from the ordered Gerbers and drill files (copper geometry only, net names ignored) also found 149/149 nets, 372/372 pins, 0 opens and 0 shorts, and it detected a deliberately injected swap of two U1 pins as 2 opens [F: project log].

8 Manufacturing outputs

The fabrication package in fab/ was generated on 2026-09-17 (KiCad 10.0.6) and ordered as a bare board: 100 × 100 mm, 2 layers, 1.6 mm, HASL, no assembly. The automated manufacturability check for JLCPCB reported READY [F].

Table 21. Gerber and drill set [F].

File	Function	Size (bytes)
tach_v1.0-F_Cu.gtl	Copper, L1, top	559 537
tach_v1.0-B_Cu.gbl	Copper, L2, bottom	527 721
tach_v1.0-F_Mask.gts / -B_Mask.gbs	Solder mask top / bottom (negative)	14 241 / 14 241

File	Function	Size (bytes)
tach_v1.0-F_Silkscreen.gto / -B_Silkscreen.gbo	Legend top / bottom	222 771 / 44 623
tach_v1.0-Edge_Cuts.gm1	Profile	1 040
tach_v1.0-PTH.drl	Excellon, plated: 7 tools, 388 holes	6 675
tach_v1.0-NPTH.drl	Excellon, non-plated: 1 tool, 4 holes	409
tach_v1.0-job.gbrjob	Gerber job file (stack-up, rules)	2 566
tach_v1.0_gerbers_jlcpcb.zip	the 10 files above, as uploaded	293 113

Plated and non-plated holes are in separate drill files, and an older single drill file without the mounting holes was deleted so it could not reach the fabricator. The 3D models attached on 2026-09-23 changed the board file but not the copper: tracks and vias (956 items), pour areas (to 0.0001 mm²), pads (9 moved by 1 nm, float rounding), masks, silkscreen and outline (within 1 µm) and the drill files were compared with the pre-change snapshot and found identical. The Gerbers were therefore not regenerated, and fab/ remains exactly what was ordered [F]. Also in fab/: the BOM (Section 9), an assembly PDF, the labelled schematic PDF, the fitted-parts assembly map (Figure 21) and 3D previews.

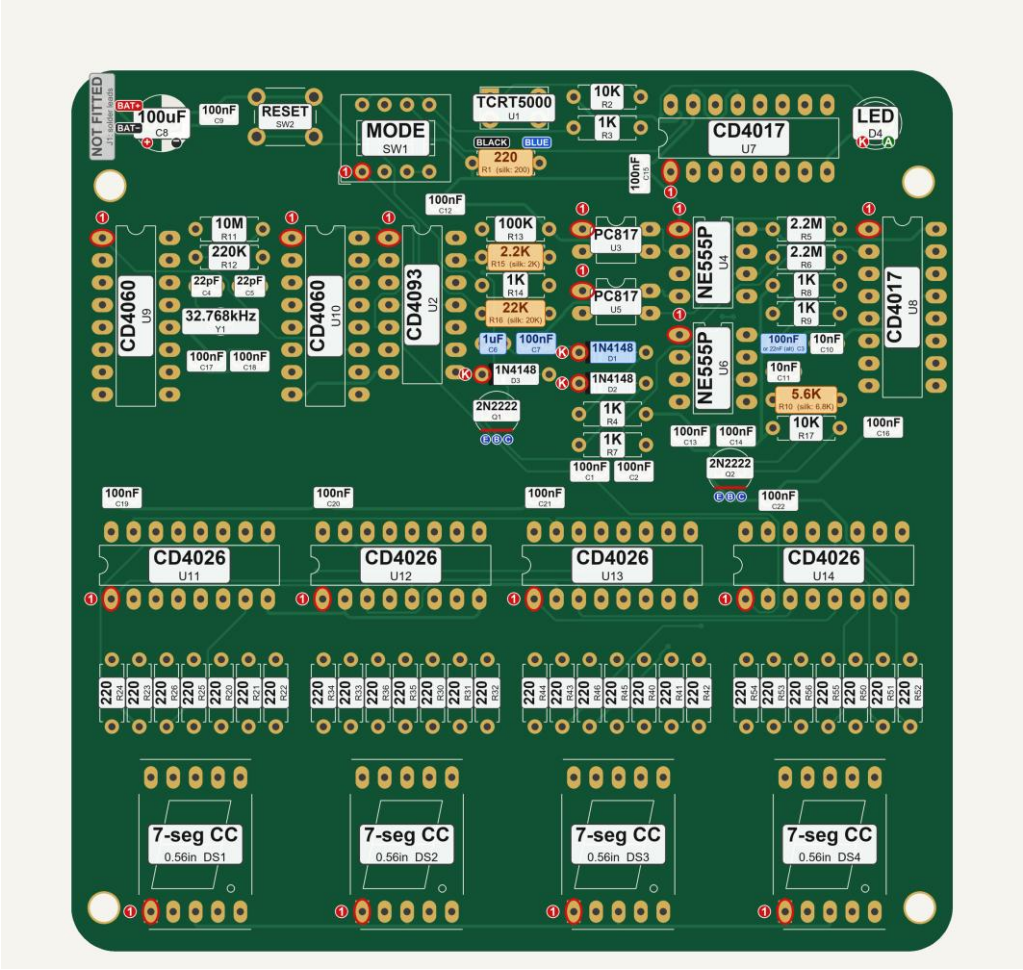


Figure 21. Assembly map with the value actually fitted at each position (orange: fitted value differs from the silkscreen; blue: no value printed on the board). Note: this map predates the TCRT5000 correction; the two IR-LED legs must be crossed (Section 10.1).

9 Bill of materials and cost

9.1 Summary

Table 22. Component summary, 95 components [F].

Group	Parts	Qty
ICs	CD4026 ×4 (U11–U14), CD4060 ×2 (U9, U10), CD4017 ×2 (U7, U8), CD4093 (U2), NE555P ×2 (U4, U6), PC817 ×2 (U3, U5)	13
Sensor	TCRT5000 (U1), custom no-clip footprint	1
Semiconductors	2N2222 TO-92 (Q1, Q2), 1N4148 (D1–D3), 5 mm LED (D4)	6
Displays	0.56-inch red common-cathode 7-segment (DS1–DS4)	4
Resistors	45 × ¼ W: 28 × 220 Ω segment, 220 Ω (R1), 1 kΩ ×6, 10 kΩ ×2, 2.2 MΩ ×2, 5.6 kΩ, 10 MΩ, 220 kΩ,	45

Group	Parts	Qty
	100 kΩ, 2.2 kΩ, 22 kΩ	
Capacitors	100 nF ×16, 22 pF ×2, 10 nF ×2, 1 µF ×1, 100 µF ×1	22
Other	32.768 kHz crystal (Y1), 4-way DIP switch (SW1), 6 mm push button (SW2), power input J1 (not fitted)	4
Mechanical	IC sockets DIP-16 ×8, DIP-14 ×1, DIP-8 ×4 (two cut to 4 positions for the PC817s); M3 hardware	—

9.2 Fitted values that differ from the drawing

Table 23. Fitted-value deviations [F: fab/tach_v1.0_bom_updated.csv].

Ref	Drawn / silkscreen	Fitted	Function	Effect
R1	200 Ω	220 Ω	TCRT5000 IR-LED current	I_F 24.0 → 21.8 mA
R10	6.8 kΩ	5.6 kΩ	Q2 base resistor	more base drive; Q2 only needs to saturate
R15	2 kΩ	2.2 kΩ	Q1 base resistor	negligible
R16	20 kΩ	22 kΩ	/CNT_CLK pull-down	negligible
C6	1 µF ceramic (BOM)	1 µF electrolytic	reset coupling	acceptable, + on /GATE_SEL (Section 4.5)
J1	2-pin header	not fitted	power input	battery leads soldered into the J1 holes

C3 is 100 nF in the schematic, on the silkscreen and on both built boards. A 22 nF part was bought as a planned alternative (Section 5.4) [F].

9.3 Cost

Table 24. Parts cost for one board. Unit prices: TechShopBD retail listings, September 2026, rounded; PCB ≈ 800 BDT; miscellaneous is an estimate [F, A].

Item	Qty	Unit (BDT)	Total (BDT)
TCRT5000 reflective optical sensor	1	15	15
CD4026	4	40	160
CD4060	2	25	50
CD4017	2	20	40
CD4093	1	20	20
NE555	2	15	30
PC817	2	5	10
2N2222 (TO-92)	2	4	8
1N4148	3	1.5	4.5
Crystal 32.768 kHz	1	15	15
7-segment display 0.56-inch CC	4	10	40
LED 5 mm	1	1.5	1.5
Resistors ¼ W	45	0.5	22.5
Ceramic capacitors	21	1.5	31.5
Electrolytic 100 µF	1	3	3
DIP switch, 4-way	1	15	15
Push button 6 mm	1	2.5	2.5
IC sockets	13	5	65
PCB fabrication (2-layer, 100 × 100 mm)	1	800	800
Battery holder 4 × AA	1	70	70
AA alkaline cell	4	30	120
Miscellaneous: reflective tape, M3 hardware, wire (estimate)	1	50	50
Total			**1,573.5**

9.4 Sourcing notes

- Parts whose counterfeits commonly misbehave (crystal, CD4060, CD4026, PC817, TCRT5000) were flagged for purchase from an authorised distributor (LCSC); everything else was bought locally [F].
- Buy the 2N2222 in the TO-92 plastic package. One local listing specifies the TO-18 metal can, whose triangular leads do not fit the inline footprint. On the footprint the square pad is the emitter and the middle pad the base; check the E-B-C order of the actual part with a meter's diode test [F].
- DIP-4 sockets are rarely stocked. The PC817 footprints have 7.62 mm row spacing, so a DIP-8 socket cut down to 4 positions fits [F].
- The displays must be common-cathode; a common-anode part will not light [F].

10 Assembly and staged bring-up

10.1 Assembly notes

- **TCRT5000 orientation (mandatory).** Because of the symbol error (Section 11), a TCRT5000 fitted straight into the footprint has its IR-LED cathode on pad 3 (which the board connects to R1) and its anode on pad 4 (ground), so the LED is reverse-biased. Fit the black-lens (detector) legs straight into the square pad (pad 1, collector) and the pad below it (pad 2, emitter), and cross the two blue-lens (LED) legs: anode into pad 3 (bottom-right, connected to R1), cathode into pad 4 (top-right, ground). Identify the anode with a meter diode test (about 1.0–1.2 V forward) and insulate one leg so the crossed legs cannot touch [F, O].
- **TCRT5000 locating pegs.** The part has two plastic pegs that sat in 2.5 mm holes in KiCad's standard footprint; the no-clip footprint has no such holes, so the pegs are cut flush. The four soldered legs hold the part [F].
- **Sensor direction.** The sensor and the displays both face up, so the display faces the target while measuring. Each reading is held for 3 s, which allows aiming and then turning the board to read; a sensor on a short 4-wire lead is more comfortable [F].
- **Polarised parts.** C8: long lead in the + pad (left). C6 (if electrolytic): + on pad 1 (/GATE_SEL). D1–D3: band on the ringed pad. D4: short lead in the square pad. Displays: decimal point bottom-right; pin 1 is the square pad [F].
- **Resistors.** Measure before fitting: 10 MΩ/1 MΩ and 2.2 MΩ/2.2 kΩ are easily confused [F].
- **Never touch a bare battery lead to the board.** Board 1 was damaged this way (Section 11). Use a 1 kΩ series lead to force a node high during tests [F].

10.2 Staged bring-up procedure

Board 2 was built one functional block at a time with only a multimeter, a series-resistor LED probe and a 1 kΩ pull-up lead; each stage is tested before the next is fitted. The procedure is generated from tools/buildguide/stages.py (11 stages, all 95 parts assigned exactly once) [F].

Table 25. Bring-up stages, test points and expected readings [F: tools/buildguide].

Stage	Fit / insert	Test point	Expected (working board)
0 Bare board	—	J1 pin 1 to pin 2	open circuit (no beep)
1 Power	J1 leads, C8, C9, C12–C22, 13 sockets	14 socket V_{DD} pins; 21 socket ground pins	V_{bat} on each; continuity to GND on each
2 Oscillator	Y1, C4, C5, R11, R12, U9	U9 pin 3 (Q14, 2 Hz); U9 pin 1 (8 Hz)	LED probe blinks 2×/s; ≈ ½ V_{bat} on a meter
3 Divider	U10	U10 pin 7; U10 pin 5	toggles every 1 s; every 2 s (10 cycles = 40 s)
4 Window logic	R2, R13, SW1, U2	U2 pins 3, 10, 4, 11; SW1 pin 7	pin 10 high 1 s in 4 s; pin 11 low during window with SW1-2 on
5 Sensor	R1, U1 (legs crossed), R3, D4	U1 anode pad; U1 pad 1; U2 pin 4	1.1–1.3 V; V_{bat} dark, < 1 V with white card at 2–10 mm; D4 follows
6 Counters, displays	R20–R56, DS1–DS4, R14, C7, D3, SW2, R16, U11–U14	segment paths; display in CNT mode	220 Ω each socket-to-display; counts 0001…0012 per card pass
7 RPS mode	Q1, R15, C6	display; U14 pin 1	blank 1 s in 4 s; ≈ V_{bat} − 1 V on U14 pin 1 during the window
8 Burst oscillator	R8, R9, C3, C11, U6	U6 pin 2; U6 pin 3	≈ ½ V_{bat} (average of the 4.8 kHz waveform); 2–4.5 V average
9 Latch	C1, D1, R4, R5, R6, C10, U3, U4	U4 pins 2, 6, 3	pin 3 goes high on a card pass and stays; low when pin 6 is pulled up
10 RPM mode	R10, Q2, R17, C2, D2, R7, U5, U7, U8	display in RPM mode	one pass in the window reads 0060, two passes 0120

11 Debugging record

Table 26. Fault log [F: project log; O].

Date	Symptom	Diagnosis / root cause	Action and result
2026-09-16	(design) segment resistors 470 Ω on the first schematic	CD4026 limits segment current to ≈ 1.6–3.2 mA at 5 V; 470 Ω wastes headroom	all 28 changed to 220 Ω before fabrication
2026-09-24	Board 1 first power-up: display stuck at 0; no IR visible on a phone camera	TCRT5000 symbol names pad 3 A and pad 4 K; datasheet top view and footprint geometry give pad 3 = cathode, pad 4 = anode. R1 fed the cathode and the anode was grounded, so the LED was reverse-biased	LED legs swapped (crossed); counter mode then counted and reset correctly (2026-09-25)

Date	Symptom	Diagnosis / root cause	Action and result
2026-09-24	DS1 and DS3 missing segment f in counter mode	suspected open in the segment-f path of those digits (R25 / DS1 pin 9 / U11 pin 6; R45 / DS3 pin 9 / U13 pin 6); DS2 and DS4 correct, so the pin-out is right	not resolved before board 1 was damaged; stage 6 of the board-2 procedure tests every segment path before the ICs are inserted
2026-09-25	Board 1: RPM and RPS modes not working after the sensor fix	blocks not exercised by counter mode: time base, window logic, SW1-2, Q1, C6 reset	diagnosis stopped when board 1 was damaged; root cause not established
2026-09-25	(verification) is the copper right?	copper and drill rebuilt from the Gerbers and compared with the schematic: 149/149 nets, 372/372 pins, 0 opens, 0 shorts	PCB confirmed; only the TCRT5000 LED polarity is wrong
2026-09-26	Board 1 damaged	a bare battery wire touched a display pin	board 2 built stage by stage (Table 25); working, RPM readings close to the reference

TCRT5000 pin evidence. Vishay datasheet (p. 1 top view and p. 4 “footprint top view”): collector top-left, anode top-right, emitter bottom-left, cathode bottom-right. The KiCad footprint has the same geometry, with the φ1 locating hole beside the emitter. By pad position this gives 1 = C, 2 = E, 3 = K, 4 = A, while tach_lib has 3 = A and 4 = K. Only the LED half is affected; the detector half (1 = C, 2 = E) is correct [D, F].

12 Experimental validation

Setup. Board 2, RPM mode, 4 × AA supply. The target was a PC case fan with a single strip of reflective tape on the hub; the TCRT5000 was held 30 mm from the fan. The reference instrument was a Lutron DT-2236 photo/contact tachometer measuring the same fan [O].

Result. The OptoTACH readings were close to those of the DT-2236 [O]. The individual readings were not recorded in the project files, so no numerical comparison is given here; Table 27 is the template for recording them. For a fan in the typical 1 000–2 000 RPM range the expected agreement is limited by quantisation to less than one count (60 RPM, 3–6 %) per reading (Section 5.3). The 30 mm working distance is twice the 15 mm limit of the datasheet response curve, which was measured against a different reference surface; the reflective tape makes this possible.

Table 27. Comparison with the reference instrument (to be filled with measured data).

Test	DT-2236 (RPM)	OptoTACH (RPM)	Difference (RPM)	Difference (%)
1				
2				
3				
4				
5				

Recommended further tests. (1) Inject a known square wave at /PULSE (signal generator, 1–100 Hz) to separate counting error from sensor error; 50 Hz must read 3000 RPM. (2) Find the board's actual ceiling by stepping in 1 Hz steps around 75–85 Hz and noting where the reading halves, or measure f at U6 pin 3. (3) Characterise the working distance and ambient-light sensitivity with the tape and with white paper.

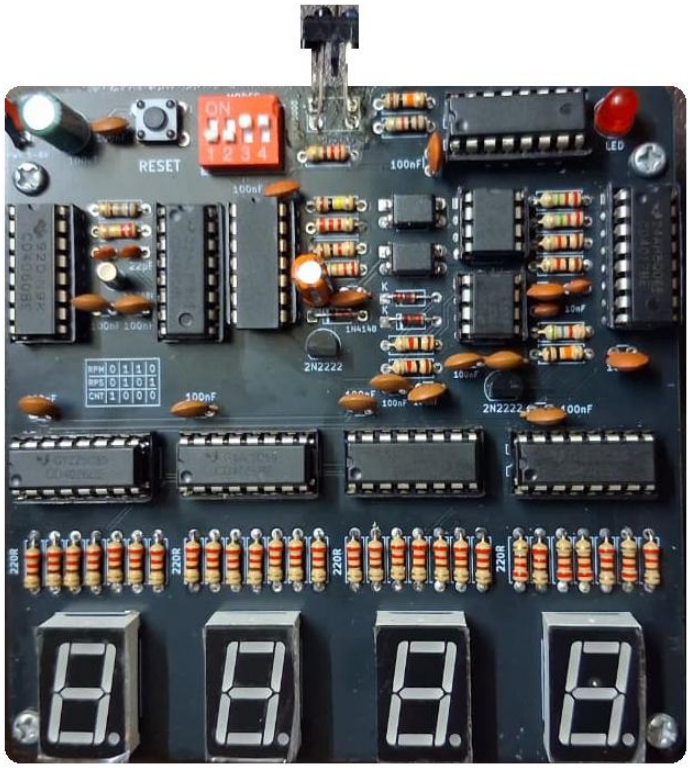


Figure 22. Assembled OptoTACH board (photograph; background removed).

13 Known issues, errata and design margins

Table 28. Errata and margins.

#	Item	Consequence	Status / workaround
1	TCRT5000 symbol pins 3 (A) and 4 (K) swapped	IR LED reverse-connected as built	legs crossed at assembly; fix in v1.1
2	Reset network loads a CMOS output with 1 kΩ through 1 µF	reset peak ≈ 2.8 V typ at 5 V, below guaranteed V_{IH}	works on both boards; v1.1: C6 100 nF, R14 10 kΩ
3	RPM ceiling 4,809 RPM with C3 = 100 nF	above it the reading silently halves	use below ≈ 4 700 RPM or fit C3 = 22 nF
4	U7/U8 not reset at power-on	first burst after switch-on 10–60 counts	ignore the first reading
5	SW1-3 and SW1-4 both ON	/BURST and /PULSE outputs fight	use only the three settings of Table 4
6	No display latch (CD4026)	4 s update, 1 s blank	by design; see v1.1 options
7	No sensor gain or threshold adjustment	distance and ambient-light sensitivity	reflective tape, shield from ambient IR
8	Silkscreen shows drawn values for R1, R10, R15, R16; C6 has no value label; mode table has no switch-number header	assembly confusion	use the assembly map (Figure 21); the switch body is numbered
9	PCB net names differ from the redrawn schematic	199 parity warnings	names only (Section 7.10); update PCB from schematic in v1.1
10	22 lib_footprint_mismatch, 4 lib_footprint_issues, 4 silk_overlap warnings	cosmetic	accepted
11	Sensor faces the same way as the display	display faces the target	hold reading 3 s; remote sensor in v1.1
12	TCRT5000 pegs have no holes in the no-clip footprint	pegs must be cut	cut flush; add holes in v1.1
13	DataShet/VC-829 datasheet is an active oscillator module; the board uses a passive 2-pin crystal	documentation	replace with the fitted crystal's datasheet
14	DataShet NPN datasheet covers 2N718A/2N1613 metal-can parts, not the TO-92 2N2222	E-B-C order unconfirmed from documentation	check each transistor with a meter
15	BOM note “C6 must be ceramic” was over-cautious	none	electrolytic acceptable (Section 4.5)

14 Recommended revision 1.1

Table 29. Proposed changes.

Change	Implementation	Expected effect
Correct TCRT5000 symbol	tach_lib: pin 3 = K, pin 4 = A; PCB: pad 3 → GND, pad 4 → R1	no crossed legs
Stronger reset margin	C6 = 100 nF, R14 = 10 kΩ (τ = 1 ms unchanged)	reset peak ≈ 5.0 V typ / 4.7 V min at 5 V [C]
Raise RPM ceiling	C3 = 22 nF	ceiling 21,859 RPM, above the 9 999 display limit [C, S]
Power-on state of ÷60	reset U7 and U8 from a power-on RC (e.g. through C7/R14-style network via a diode OR with U8 Q6)	first reading correct
Continuous display	segment latches (e.g. 4 × 74HC573) clocked at window end, or CD40110 if obtainable	update every 1–4 s without blanking
Better front end	comparator with adjustable threshold and a gain stage ahead of U2B	wider distance range, ambient immunity
Safer mode selection	rotary switch or series resistor between /BURST and /GATED_IN	no output contention
Mechanical	sensor on a 4-wire connector; TCRT peg holes; silkscreen with fitted values and switch numbers	easier use and assembly
Documentation	update PCB from schematic (net names); regenerate Gerbers only after these changes	clean parity report

15 Project files and reproducibility

Table 30. Repository layout.

Path	Content
tach_v1.0.kicad_pro / .kicad_sch / .kicad_pcb	KiCad 10 project, schematic, board
tach_v1.0.kicad_dru	custom DRC rule (GND thermal spokes)
tach_lib.kicad_sym, sym-lib-table	custom symbols CD4026, TCRT5000
tach_fp.pretty/, fp-lib-table	custom footprint TCRT5000_NoClip
fab/	Gerbers and drill (as ordered), zip, BOM (bom.csv, bom_updated.csv), purchase list, assembly PDF, schematic PDF, fitted-parts map, 3D previews (.png, .glb), SHOPPING_LIST.md
analysis/	accuracy report, behavioural simulation (sweep.py), results (sweep.json), figures, exported netlist
routing/	Specctra DSN/SES files, Freerouting logs, pipeline scripts (export_dsn, patch_dsn, protect_wiring, finalize_board, diff_net_tracks)
tools/bom/	make_bom.py (BOM with C3 alternate and fitted values), purchase_pdf.py
tools/silkscreen/	label solver and audit scripts (labels, retight, fillgaps, audit_silk, coverage)
tools/logo/	geometric typeface, SVG flattener, optotach_flat.svg (back-side artwork)
tools/placement/	mounting-hole and clear-area search, netlist query helper
tools/sch_wiring/	grid maze router that converted the labelled schematic to wires; netlist comparison
tools/fitmap/	fitted-parts map generator and BOM cross-check
tools/buildguide/	staged bring-up plan (stages.py), stage maps, HTML guide
snapshots/	zipped known-good states (2026-09-16 … 2026-09-23) and the user-edited BOM
DataShet/	datasheets used (see Section 13 for two mismatches)
tach_v1.0_for_sharing.zip	minimal set to open the design on another machine (needs KiCad 10)

Toolchain. KiCad 10.0.6 with kicad-cli; KiCad's bundled Python (pcbnew module) for board analysis; Freerouting for autorouting; a KiCad IPC automation server (Konnect) for scripted schematic and board edits; Python 3 with matplotlib for analysis figures.

Reproduce the checks.

- `kicad-cli sch erc --severity-all tach_v1.0.kicad_sch`
- `kicad-cli pcb drc --severity-all --schematic-parity tach_v1.0.kicad_pcb`
- `kicad-cli sch export netlist --format kicadsexpr -o tach.net tach_v1.0.kicad_sch`
- `python analysis/sweep.py` (writes sweep.json; parameters R8, R9, C3, BURST_N, GATE_LEN at the top of the file)
- `kicad-cli pcb render --side top --quality high -o top.png tach_v1.0.kicad_pcb`

KiCad has no merge: two people editing the same file lose one set of changes. Keep the library tables and custom libraries in the project folder (they use ${KIPRJMOD}).

16 Conclusions

OptoTACH shows that an accurate, direct-reading RPM meter can be built from standard CMOS logic without a microcontroller. Its accuracy rests on two properties that need no calibration: the counting window is derived from a quartz crystal by binary division alone, and the ×60 multiplication is performed by counting, not timing. The substitution of the unavailable CD40110 by the CD4026 was solved with gates already on the board, at the cost of a 4 s update interval. The design was verified at every level available: electrical rules, design rules, an explicit pin-by-pin connectivity proof between schematic and copper, an independent Gerber-level check, timing analysis and a behavioural simulation. Bring-up exposed one design error (TCRT5000 LED polarity in the custom symbol), and the analysis in this report identified the limits users must know (60 RPM resolution with one mark, a silent half-reading above 4 809 RPM with C3 = 100 nF) and three design-margin items. All are documented with corrections for a revision 1.1. The second board works, and its RPM readings were close to a commercial tachometer; recording those readings numerically is the next step for a publication.

References

[1] Mousa Simple Projects, “Non Contact Tachometer Without Microcontroller,” mousa-simple-project.com, 2020 (reference topology; parts list reproduced in Table 1).

[2] Vishay Semiconductors, “TCRT5000, TCRT5000L Reflective Optical Sensor with Transistor Output,” doc. 83760, rev. 1.7, 2009.

[3] Texas Instruments, “CD4026B, CD4033B CMOS Decade Counters/Dividers with Decoded 7-Segment Display Outputs,” SCHS031B, 2003.

[4] Texas Instruments, “CD4060B CMOS 14-Stage Ripple-Carry Binary Counter/Divider and Oscillator.”

[5] Texas Instruments, “CD4017B, CD4022B CMOS Counter/Dividers,” SCHS027C, 2004.

[6] Texas Instruments, “CD4093B CMOS Quad 2-Input NAND Schmitt Triggers.”

[7] Texas Instruments, “NA555, NE555, SA555, SE555 Precision Timers,” SLFS022K, rev. March 2026.

[8] Sharp, “PC817XxNSZ1B Series Photocoupler,” sheet OP18002EN.

[9] onsemi, “1N914/1N4148 Small Signal Diode,” 1N914-D.

[10] Lutron Electronic, “DT-2236 Photo/Contact Tachometer,” operation manual.

[11] KiCad EDA 10.0 documentation: kicad-cli, custom design rules (.kicad_dru).

Appendix A Complete bill of materials with placement

Table 31. All 99 footprints ("=" fitted as drawn). Coordinates in mm, KiCad absolute [F].

Ref	Value (drawn)	Fitted	Footprint	X	Y	Rot
C1	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	157.60	95.50	0
C2	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	162.60	95.50	0
C3	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	179.60	81.00	0
C4	22pF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	114.10	74.50	0
C5	22pF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	119.00	74.50	0
C6	1uF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	146.60	81.00	0
C7	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	151.50	81.00	0
C8	100uF	=	CP_Radial_D6.3mm_P2.50mm	109.00	56.00	0
C9	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	115.50	55.00	0
C10	10nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	184.50	81.00	0
C11	10nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	179.60	84.20	0
C12	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	141.20	65.20	0
C13	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	169.00	91.50	0
C14	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	174.20	91.50	0
C15	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	164.50	63.00	90
C16	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	190.90	90.50	0
C17	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	114.10	83.00	0
C18	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	119.00	83.00	0
C19	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	104.50	98.50	0
C20	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	128.50	98.50	0
C21	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	152.50	98.50	0
C22	100nF	=	C_Disc_D3.0mm_W1.6mm_P2.50mm	179.00	98.80	0
D1	1N4148	=	D_DO-35_SOD27_P7.62mm_Horizontal	157.60	82.00	0
D2	1N4148	=	D_DO-35_SOD27_P7.62mm_Horizontal	157.60	85.50	0
D3	1N4148	=	D_DO-35_SOD27_P7.62mm_Horizontal	146.60	84.50	0
D4	LED	=	LED_D5.0mm	190.00	56.00	0
DS1	7SEG_CC_0.56in	=	7SegmentLED_LTS6760_LTS6780	109.00	145.50	90
DS2	7SEG_CC_0.56in	=	7SegmentLED_LTS6760_LTS6780	133.00	145.50	90
DS3	7SEG_CC_0.56in	=	7SegmentLED_LTS6760_LTS6780	157.00	145.50	90
DS4	7SEG_CC_0.56in	=	7SegmentLED_LTS6760_LTS6780	181.00	145.50	90
H1	MountingHole 3.2 mm	—	MountingHole_3.2mm	104.38	63.08	0
H2	MountingHole 3.2 mm	—	MountingHole_3.2mm	196.13	62.67	0
H3	MountingHole 3.2 mm	—	MountingHole_3.2mm	103.68	145.07	0
H4	MountingHole 3.2 mm	—	MountingHole_3.2mm	196.28	145.07	0
J1	PWR 5-6V	not fitted	PinHeader_1x02_P2.54mm_Vertical	103.50	54.00	0
Q1	2N2222	=	TO-92_Inline	147.00	89.00	0
Q2	2N2222	=	TO-92_Inline	173.50	95.50	0
R1	200	220	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	145.50	60.50	0
R2	10K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	157.00	53.00	0
R3	1K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	157.00	56.50	0
R4	1K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	157.60	89.00	0
R5	2.2M	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	179.60	68.00	0
R6	2.2M	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	179.60	71.20	0
R7	1K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	157.60	92.50	0
R8	1K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	179.60	74.40	0
R9	1K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	179.60	77.60	0
R10	6.8K	5.6K	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	179.60	87.40	0
R11	10M	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	114.10	68.00	0
R12	220K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	114.10	71.20	0
R13	100K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	146.60	68.00	0
R14	1K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	146.60	74.40	0
R15	2K	2.2K	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	146.60	71.20	0
R16	20K	22K	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	146.60	77.60	0
R17	10K	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	179.60	90.60	0
R20	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	117.00	124.50	90
R21	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	120.10	124.50	90
R22	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	123.20	124.50	90
R23	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	107.70	124.50	90
R24	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	104.60	124.50	90
R25	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	113.90	124.50	90
R26	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	110.80	124.50	90
R30	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	141.00	124.50	90
R31	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	144.10	124.50	90
R32	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	147.20	124.50	90
R33	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	131.70	124.50	90
R34	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	128.60	124.50	90
R35	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	137.90	124.50	90
R36	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	134.80	124.50	90
R40	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	165.00	124.50	90
R41	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	168.10	124.50	90
R42	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	171.20	124.50	90

Ref	Value (drawn)	Fitted	Footprint	X	Y	Rot
R43	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	155.70	124.50	90
R44	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	152.60	124.50	90
R45	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	161.90	124.50	90
R46	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	158.80	124.50	90
R50	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	189.00	124.50	90
R51	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	192.10	124.50	90
R52	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	195.20	124.50	90
R53	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	179.70	124.50	90
R54	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	176.60	124.50	90
R55	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	185.90	124.50	90
R56	220	=	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	182.80	124.50	90
SW1	MODE	=	SW_DIP_SPSTx04_Slide_9.78x12.34mm_W7.62mm_P2.54mm	133.00	61.50	90
SW2	RESET	=	SW_PUSH_6mm	121.00	53.00	0
U1	TCRT5000	=	TCRT5000_NoClip	147.50	53.00	0
U2	CD4093	=	DIP-14_W7.62mm_LongPads	136.00	69.00	0
U3	PC817	=	DIP-4_W7.62mm_LongPads	158.00	68.00	0
U4	NE555P	=	DIP-8_W7.62mm_LongPads	169.00	68.00	0
U5	PC817	=	DIP-4_W7.62mm_LongPads	158.00	75.00	0
U6	NE555P	=	DIP-8_W7.62mm_LongPads	169.00	80.00	0
U7	CD4017	=	DIP-16_W7.62mm_LongPads	168.00	61.50	90
U8	CD4017	=	DIP-16_W7.62mm_LongPads	190.50	68.00	0
U9	CD4060	=	DIP-16_W7.62mm_LongPads	103.50	69.00	0
U10	CD4060	=	DIP-16_W7.62mm_LongPads	125.00	69.00	0
U11	CD4026	=	DIP-16_W7.62mm_LongPads	104.50	110.00	90
U12	CD4026	=	DIP-16_W7.62mm_LongPads	128.50	110.00	90
U13	CD4026	=	DIP-16_W7.62mm_LongPads	152.50	110.00	90
U14	CD4026	=	DIP-16_W7.62mm_LongPads	176.50	110.00	90
Y1	32.768kHz	=	Crystal_C38-LF_D3.0mm_L8.0mm_Vertical	116.00	78.50	0

Appendix B Netlist

All 96 multi-pin nets. PCB name, the automatic name the redrawn schematic assigns to the same pins, pin count and members (reference.pin). /GND (58 pins) and /VCC (35 pins) are listed in Appendix D per IC and in Section 4.1 [F].

Table 32. Multi-pin nets [F].

PCB net	Schematic net	Pins	Members
/BURST	Net-(Q2-C)	4	Q2.3 R17.2 SW1.6 U7.14
/BURST_DONE	Net-(U8-Q1)	2	C2.1 U8.2
/CARRY_HUNDREDS	Net-(U11-CLK)	2	U11.1 U12.5
/CARRY_TENS	Net-(U12-CLK)	2	U12.1 U13.5
/CARRY_UNITS	Net-(U13-CLK)	2	U13.1 U14.5
/CLK_8HZ	Net-(U10-Φ1)	2	U9.1 U10.11
/CNT_CLK	Net-(Q1-E)	4	Q1.1 R16.1 SW1.1 U14.1
/CO10	Net-(U7-Cout)	2	U7.12 U8.14
/CV1	Net-(U4-CONT)	2	C10.1 U4.5
/CV2	Net-(U6-CONT)	2	C11.1 U6.5
/DISP_EN	Net-(U11-DEI)	5	U2.11 U11.3 U12.3 U13.3 U14.3
/DIV6_RST	Net-(U8-Q6)	2	U8.5 U8.15
/GATE	Net-(SW1-Pad2)	2	SW1.2 U2.10
/GATED_IN	Net-(Q1-C)	3	Q1.3 SW1.3 SW1.4
/GATE_N	Net-(U2-Pad3)	3	U2.3 U2.8 U2.9
/GATE_SEL	Net-(C6-Pad1)	6	C6.1 R13.1 R15.1 SW1.7 U2.12 U2.13
/GND	GND	58	see text
/IR_LED_A	Net-(U1-A)	2	R1.2 U1.3
/LATCH_Q	Net-(U4-OUT)	2	R10.1 U4.3
/LED1_a	Net-(DS1-a)	2	DS1.7 R20.2
/LED1_b	Net-(DS1-b)	2	DS1.6 R21.2
/LED1_c	Net-(DS1-c)	2	DS1.4 R22.2
/LED1_d	Net-(DS1-d)	2	DS1.2 R23.2
/LED1_e	Net-(DS1-e)	2	DS1.1 R24.2
/LED1_f	Net-(DS1-f)	2	DS1.9 R25.2
/LED1_g	Net-(DS1-g)	2	DS1.10 R26.2
/LED2_a	Net-(DS2-a)	2	DS2.7 R30.2
/LED2_b	Net-(DS2-b)	2	DS2.6 R31.2
/LED2_c	Net-(DS2-c)	2	DS2.4 R32.2
/LED2_d	Net-(DS2-d)	2	DS2.2 R33.2
/LED2_e	Net-(DS2-e)	2	DS2.1 R34.2
/LED2_f	Net-(DS2-f)	2	DS2.9 R35.2
/LED2_g	Net-(DS2-g)	2	DS2.10 R36.2
/LED3_a	Net-(DS3-a)	2	DS3.7 R40.2
/LED3_b	Net-(DS3-b)	2	DS3.6 R41.2
/LED3_c	Net-(DS3-c)	2	DS3.4 R42.2
/LED3_d	Net-(DS3-d)	2	DS3.2 R43.2
/LED3_e	Net-(DS3-e)	2	DS3.1 R44.2
/LED3_f	Net-(DS3-f)	2	DS3.9 R45.2
/LED3_g	Net-(DS3-g)	2	DS3.10 R46.2
/LED4_a	Net-(DS4-a)	2	DS4.7 R50.2
/LED4_b	Net-(DS4-b)	2	DS4.6 R51.2

PCB net	Schematic net	Pins	Members
/LED4_c	Net-(DS4-c)	2	DS4.4 R52.2
/LED4_d	Net-(DS4-d)	2	DS4.2 R53.2
/LED4_e	Net-(DS4-e)	2	DS4.1 R54.2
/LED4_f	Net-(DS4-f)	2	DS4.9 R55.2
/LED4_g	Net-(DS4-g)	2	DS4.10 R56.2
/LED_IND	Net-(D4-A)	2	D4.2 R3.2
/OSC	Net-(Q2-E)	2	Q2.1 U6.3
/OSC_DIS	Net-(U6-DISCH)	3	R8.2 R9.1 U6.7
/OSC_TH	Net-(U6-THRES)	4	C3.1 R9.2 U6.2 U6.6
/PULSE	Net-(C1-Pad1)	5	C1.1 R3.1 SW1.5 SW1.8 U2.4
/Q1_B	Net-(Q1-B)	2	Q1.2 R15.2
/Q2_B	Net-(Q2-B)	2	Q2.2 R10.2
/RESET	Net-(D3-K)	9	C6.2 C7.2 D3.1 R14.1 SW2.2 U11.15 U12.15 U13.15 U14.15
/RST_A	Net-(D2-K)	3	C2.2 D2.1 U5.1
/RST_K	Net-(D2-A)	3	D2.2 R7.1 U5.2
/SEG1_a	Net-(U11-a)	2	R20.1 U11.10
/SEG1_b	Net-(U11-b)	2	R21.1 U11.12
/SEG1_c	Net-(U11-c)	2	R22.1 U11.13
/SEG1_d	Net-(U11-d)	2	R23.1 U11.9
/SEG1_e	Net-(U11-e)	2	R24.1 U11.11
/SEG1_f	Net-(U11-f)	2	R25.1 U11.6
/SEG1_g	Net-(U11-g)	2	R26.1 U11.7
/SEG2_a	Net-(U12-a)	2	R30.1 U12.10
/SEG2_b	Net-(U12-b)	2	R31.1 U12.12
/SEG2_c	Net-(U12-c)	2	R32.1 U12.13
/SEG2_d	Net-(U12-d)	2	R33.1 U12.9
/SEG2_e	Net-(U12-e)	2	R34.1 U12.11
/SEG2_f	Net-(U12-f)	2	R35.1 U12.6
/SEG2_g	Net-(U12-g)	2	R36.1 U12.7
/SEG3_a	Net-(U13-a)	2	R40.1 U13.10
/SEG3_b	Net-(U13-b)	2	R41.1 U13.12
/SEG3_c	Net-(U13-c)	2	R42.1 U13.13
/SEG3_d	Net-(U13-d)	2	R43.1 U13.9
/SEG3_e	Net-(U13-e)	2	R44.1 U13.11
/SEG3_f	Net-(U13-f)	2	R45.1 U13.6
/SEG3_g	Net-(U13-g)	2	R46.1 U13.7
/SEG4_a	Net-(U14-a)	2	R50.1 U14.10
/SEG4_b	Net-(U14-b)	2	R51.1 U14.12
/SEG4_c	Net-(U14-c)	2	R52.1 U14.13
/SEG4_d	Net-(U14-d)	2	R53.1 U14.9
/SEG4_e	Net-(U14-e)	2	R54.1 U14.11
/SEG4_f	Net-(U14-f)	2	R55.1 U14.6
/SEG4_g	Net-(U14-g)	2	R56.1 U14.7
/SENS_RAW	Net-(U1-C)	4	R2.2 U1.1 U2.5 U2.6
/SET_A	Net-(D1-K)	3	C1.2 D1.1 U3.1
/SET_K	Net-(D1-A)	3	D1.2 R4.1 U3.2
/THRES	Net-(U4-THRES)	3	R6.1 U4.6 U5.3
/TRIG	Net-(U4-TRIG)	3	R5.2 U3.4 U4.2
/T_Q4_0.5HZ	Net-(U10-Q4)	2	U2.1 U10.7
/T_Q5_0.25HZ	Net-(U10-Q5)	2	U2.2 U10.5
/VCC	VCC	35	see text
/XTAL_DRV	Net-(U9-Φ0)	3	R11.2 R12.1 U9.10
/XTAL_IN	Net-(U9-Φ1)	4	C4.1 R11.1 U9.11 Y1.1
/XTAL_OUT	Net-(C5-Pad1)	3	C5.1 R12.2 Y1.2

Appendix C Unconnected pins

Table 33. The 53 pins with a no-connect flag [F].

Ref	Unconnected pins
DS1	5
DS2	5
DS3	5
DS4	5
U4	7
U7	1, 2, 3, 4, 5, 6, 7, 9, 10, 11
U8	1, 3, 4, 6, 7, 9, 10, 11, 12
U9	2, 3, 4, 5, 6, 7, 9, 13, 14, 15
U10	1, 2, 3, 4, 6, 9, 10, 13, 14, 15
U11	4, 5, 14
U12	4, 14
U13	4, 14
U14	4, 14

Appendix D IC pin assignments

Table 34. U1 — TCRT5000 [F].

Pin	Function	Net	Pin	Function	Net
1	collector	/SENS_RAW	3	“A” in symbol (physically cathode)	/IR_LED_A
2	emitter	/GND	4	“K” in symbol (physically anode)	/GND

Table 35. U2 — CD4093 [F].

Pin	Function	Net	Pin	Function	Net
1	A1	/T_Q4_0.5HZ	8	A3	/GATE_N
2	B1	/T_Q5_0.25HZ	9	B3	/GATE_N
3	Y1	/GATE_N	10	Y3	/GATE
4	Y2	/PULSE	11	Y4	/DISP_EN
5	A2	/SENS_RAW	12	A4	/GATE_SEL
6	B2	/SENS_RAW	13	B4	/GATE_SEL
7	VSS	/GND	14	VDD	/VCC

Table 36. U3 — PC817 [F].

Pin	Function	Net	Pin	Function	Net
1	anode	/SET_A	3	emitter	/GND
2	cathode	/SET_K	4	collector	/TRIG

Table 37. U4 — NE555P [F].

Pin	Function	Net	Pin	Function	Net
1	GND	/GND	5	CONT	/CV1
2	TRIG	/TRIG	6	THRES	/THRES
3	OUT	/LATCH_Q	7	DISCH	not connected
4	/RST	/VCC	8	VCC	/VCC

Table 38. U5 — PC817 [F].

Pin	Function	Net	Pin	Function	Net
1	anode	/RST_A	3	emitter	/THRES
2	cathode	/RST_K	4	collector	/VCC

Table 39. U6 — NE555P [F].

Pin	Function	Net	Pin	Function	Net
1	GND	/GND	5	CONT	/CV2
2	TRIG	/OSC_TH	6	THRES	/OSC_TH
3	OUT	/OSC	7	DISCH	/OSC_DIS
4	/RST	/VCC	8	VCC	/VCC

Table 40. U7 — CD4017 [F].

Pin	Function	Net	Pin	Function	Net
1	Q5	not connected	9	Q8	not connected
2	Q1	not connected	10	Q4	not connected
3	Q0	not connected	11	Q9	not connected
4	Q2	not connected	12	Cout	/CO10
5	Q6	not connected	13	CKEN	/GND
6	Q7	not connected	14	CLK	/BURST
7	Q3	not connected	15	Reset	/GND
8	VSS	/GND	16	VDD	/VCC

Table 41. U8 — CD4017 [F].

Pin	Function	Net	Pin	Function	Net
1	Q5	not connected	9	Q8	not connected
2	Q1	/BURST_DONE	10	Q4	not connected
3	Q0	not connected	11	Q9	not connected
4	Q2	not connected	12	Cout	not connected
5	Q6	/DIV6_RST	13	CKEN	/GND
6	Q7	not connected	14	CLK	/CO10
7	Q3	not connected	15	Reset	/DIV6_RST
8	VSS	/GND	16	VDD	/VCC

Table 42. U9 — CD4060 [F].

Pin	Function	Net	Pin	Function	Net
1	Q12	/CLK_8HZ	9	/Φ0	not connected
2	Q13	not connected	10	Φ0	/XTAL_DRV
3	Q14	not connected	11	/Φ1	/XTAL_IN

Pin	Function	Net	Pin	Function	Net
4	Q6	not connected	12	CLR	/GND
5	Q5	not connected	13	Q9	not connected
6	Q7	not connected	14	Q8	not connected
7	Q4	not connected	15	Q10	not connected
8	VSS	/GND	16	VDD	/VCC

Table 43. U10 — CD4060 [F].

Pin	Function	Net	Pin	Function	Net
1	Q12	not connected	9	/Φ0	not connected
2	Q13	not connected	10	Φ0	not connected
3	Q14	not connected	11	/Φ1	/CLK_8HZ
4	Q6	not connected	12	CLR	/GND
5	Q5	/T_Q5_0.25HZ	13	Q9	not connected
6	Q7	not connected	14	Q8	not connected
7	Q4	/T_Q4_0.5HZ	15	Q10	not connected
8	VSS	/GND	16	VDD	/VCC

Table 44. U11 — CD4026 [F].

Pin	Function	Net	Pin	Function	Net
1	CLK	/CARRY_HUNDREDS	9	d	/SEG1_d
2	INH	/GND	10	a	/SEG1_a
3	DEI	/DISP_EN	11	e	/SEG1_e
4	DEO	not connected	12	b	/SEG1_b
5	CO	not connected	13	c	/SEG1_c
6	f	/SEG1_f	14	UCS	not connected
7	g	/SEG1_g	15	RST	/RESET
8	VSS	/GND	16	VDD	/VCC

Table 45. U12 — CD4026 [F].

Pin	Function	Net	Pin	Function	Net
1	CLK	/CARRY_TENS	9	d	/SEG2_d
2	INH	/GND	10	a	/SEG2_a
3	DEI	/DISP_EN	11	e	/SEG2_e
4	DEO	not connected	12	b	/SEG2_b
5	CO	/CARRY_HUNDREDS	13	c	/SEG2_c
6	f	/SEG2_f	14	UCS	not connected
7	g	/SEG2_g	15	RST	/RESET
8	VSS	/GND	16	VDD	/VCC

Table 46. U13 — CD4026 [F].

Pin	Function	Net	Pin	Function	Net
1	CLK	/CARRY_UNITS	9	d	/SEG3_d
2	INH	/GND	10	a	/SEG3_a
3	DEI	/DISP_EN	11	e	/SEG3_e
4	DEO	not connected	12	b	/SEG3_b
5	CO	/CARRY_TENS	13	c	/SEG3_c
6	f	/SEG3_f	14	UCS	not connected
7	g	/SEG3_g	15	RST	/RESET
8	VSS	/GND	16	VDD	/VCC

Table 47. U14 — CD4026 [F].

Pin	Function	Net	Pin	Function	Net
1	CLK	/CNT_CLK	9	d	/SEG4_d
2	INH	/GND	10	a	/SEG4_a
3	DEI	/DISP_EN	11	e	/SEG4_e
4	DEO	not connected	12	b	/SEG4_b
5	CO	/CARRY_UNITS	13	c	/SEG4_c
6	f	/SEG4_f	14	UCS	not connected
7	g	/SEG4_g	15	RST	/RESET
8	VSS	/GND	16	VDD	/VCC

Appendix E Simulation data

Table 48. Behavioural simulation at every integer frequency and selected half-integer points (12 phases × 6 windows) [S].

Hz	True RPM	RPM, 100 nF	Worst %	Lost %	RPM, 22 nF	Lost %	RPS
1	60	60–60	0.00	0.0	60–60	0.0	1–1
1.5	90	60–120	33.33	0.0	60–120	0.0	1–2
2	120	120–120	0.00	0.0	120–120	0.0	2–2
2.5	150	120–180	20.00	0.0	120–180	0.0	2–3
3	180	180–180	0.00	0.0	180–180	0.0	3–3

Hz	True RPM	RPM, 100 nF	Worst %	Lost %	RPM, 22 nF	Lost %	RPS
4	240	240–240	0.00	0.0	240–240	0.0	4–4
5	300	300–300	0.00	0.0	300–300	0.0	5–5
6	360	360–360	0.00	0.0	360–360	0.0	6–6
7	420	420–420	0.00	0.0	420–420	0.0	7–7
8	480	480–480	0.00	0.0	480–480	0.0	8–8
9	540	540–540	0.00	0.0	540–540	0.0	9–9
10	600	600–600	0.00	0.0	600–600	0.0	10–10
11	660	660–660	0.00	0.0	660–660	0.0	11–11
12	720	720–720	0.00	0.0	720–720	0.0	12–12
13	780	780–780	0.00	0.0	780–780	0.0	13–13
14	840	840–840	0.00	0.0	840–840	0.0	14–14
15	900	900–900	0.00	0.0	900–900	0.0	15–15
16	960	960–960	0.00	0.0	960–960	0.0	16–16
17	1,020	1020–1020	0.00	0.0	1020–1020	0.0	17–17
18	1,080	1080–1080	0.00	0.0	1080–1080	0.0	18–18
19	1,140	1140–1140	0.00	0.0	1140–1140	0.0	19–19
20	1,200	1200–1200	0.00	0.0	1200–1200	0.0	20–20
21	1,260	1260–1260	0.00	0.0	1260–1260	0.0	21–21
22	1,320	1320–1320	0.00	0.0	1320–1320	0.0	22–22
23	1,380	1380–1380	0.00	0.0	1380–1380	0.0	23–23
24	1,440	1440–1440	0.00	0.0	1440–1440	0.0	24–24
25	1,500	1500–1500	0.00	0.0	1500–1500	0.0	25–25
26	1,560	1560–1560	0.00	0.0	1560–1560	0.0	26–26
27	1,620	1620–1620	0.00	0.0	1620–1620	0.0	27–27
28	1,680	1680–1680	0.00	0.0	1680–1680	0.0	28–28
29	1,740	1740–1740	0.00	0.0	1740–1740	0.0	29–29
30	1,800	1800–1800	0.00	0.0	1800–1800	0.0	30–30
31	1,860	1860–1860	0.00	0.0	1860–1860	0.0	31–31
32	1,920	1920–1920	0.00	0.0	1920–1920	0.0	32–32
33	1,980	1980–1980	0.00	0.0	1980–1980	0.0	33–33
34	2,040	2040–2040	0.00	0.0	2040–2040	0.0	34–34
35	2,100	2100–2100	0.00	0.0	2100–2100	0.0	35–35
36	2,160	2160–2160	0.00	0.0	2160–2160	0.0	36–36
37	2,220	2220–2220	0.00	0.0	2220–2220	0.0	37–37
38	2,280	2280–2280	0.00	0.0	2280–2280	0.0	38–38
39	2,340	2340–2340	0.00	0.0	2340–2340	0.0	39–39
40	2,400	2400–2400	0.00	0.0	2400–2400	0.0	40–40
41	2,460	2460–2460	0.00	0.0	2460–2460	0.0	41–41
42	2,520	2520–2520	0.00	0.0	2520–2520	0.0	42–42
43	2,580	2580–2580	0.00	0.0	2580–2580	0.0	43–43
44	2,640	2640–2640	0.00	0.0	2640–2640	0.0	44–44
45	2,700	2700–2700	0.00	0.0	2700–2700	0.0	45–45
46	2,760	2760–2760	0.00	0.0	2760–2760	0.0	46–46
47	2,820	2820–2820	0.00	0.0	2820–2820	0.0	47–47
48	2,880	2880–2880	0.00	0.0	2880–2880	0.0	48–48
49	2,940	2940–2940	0.00	0.0	2940–2940	0.0	49–50
50	3,000	3000–3000	0.00	0.0	3000–3000	0.0	50–50
51	3,060	3060–3060	0.00	0.0	3060–3060	0.0	51–51
52	3,120	3120–3120	0.00	0.0	3120–3120	0.0	52–52
53	3,180	3180–3180	0.00	0.0	3180–3180	0.0	53–53
54	3,240	3240–3240	0.00	0.0	3240–3240	0.0	54–54
55	3,300	3300–3300	0.00	0.0	3300–3300	0.0	55–55
56	3,360	3360–3360	0.00	0.0	3360–3360	0.0	56–56
57	3,420	3420–3420	0.00	0.0	3420–3420	0.0	57–57
58	3,480	3480–3480	0.00	0.0	3480–3480	0.0	58–58
59	3,540	3540–3540	0.00	0.0	3540–3540	0.0	59–59
60	3,600	3600–3600	0.00	0.0	3600–3600	0.0	60–60
61	3,660	3660–3660	0.00	0.0	3660–3660	0.0	61–61
62	3,720	3720–3720	0.00	0.0	3720–3720	0.0	62–62
63	3,780	3780–3780	0.00	0.0	3780–3780	0.0	63–63
64	3,840	3840–3840	0.00	0.0	3840–3840	0.0	64–64
65	3,900	3900–3900	0.00	0.0	3900–3900	0.0	65–65
66	3,960	3960–3960	0.00	0.0	3960–3960	0.0	66–66
67	4,020	4020–4020	0.00	0.0	4020–4020	0.0	67–67
68	4,080	4080–4080	0.00	0.0	4080–4080	0.0	68–68
69	4,140	4140–4140	0.00	0.0	4140–4140	0.0	69–69
70	4,200	4200–4200	0.00	0.0	4200–4200	0.0	70–70
71	4,260	4260–4260	0.00	0.0	4260–4260	0.0	71–71
72	4,320	4320–4320	0.00	0.0	4320–4320	0.0	72–72
73	4,380	4380–4380	0.00	0.0	4380–4380	0.0	73–73
74	4,440	4440–4440	0.00	0.0	4440–4440	0.0	74–74
75	4,500	4500–4500	0.00	0.0	4500–4500	0.0	75–75
76	4,560	4560–4560	0.00	0.0	4560–4560	0.0	76–76
77	4,620	4620–4620	0.00	0.0	4620–4620	0.0	77–77
78	4,680	4680–4680	0.00	0.0	4680–4680	0.0	78–78
79	4,740	4740–4740	0.00	0.0	4740–4740	0.0	79–79
79.5	4,770	4770–4770	0.00	0.0	4740–4800	0.0	79–80
80	4,800	3840–4800	20.00	6.0	4800–4800	0.0	80–80
80.5	4,830	2400–2426	50.31	50.0	4800–4860	0.0	80–81
81	4,860	2401–2456	50.60	50.0	4860–4860	0.0	81–81
82	4,920	2460–2460	50.00	50.0	4920–4920	0.0	82–82
83	4,980	2463–2516	50.54	50.0	4980–4980	0.0	83–83
84	5,040	2520–2520	50.00	50.0	5040–5040	0.0	84–84
85	5,100	2524–2576	50.51	50.0	5100–5100	0.0	85–85

Hz	True RPM	RPM, 100 nF	Worst %	Lost %	RPM, 22 nF	Lost %	RPS
86	5,160	2580–2580	50.00	50.0	5160–5160	0.0	86–86
87	5,220	2585–2635	50.48	50.0	5220–5220	0.0	87–87
88	5,280	2640–2640	50.00	50.0	5280–5280	0.0	88–88
89	5,340	2647–2694	50.43	50.0	5340–5340	0.0	89–89
90	5,400	2700–2700	50.00	50.0	5400–5400	0.0	90–90
91	5,460	2708–2753	50.40	50.0	5460–5460	0.0	91–91
92	5,520	2760–2760	50.00	50.0	5520–5520	0.0	92–92
93	5,580	2769–2812	50.38	50.0	5580–5580	0.0	93–93
94	5,640	2820–2820	50.00	50.0	5640–5640	0.0	94–94
95	5,700	2830–2870	50.35	50.0	5700–5700	0.0	95–95
96	5,760	2880–2880	50.00	50.0	5760–5760	0.0	96–96
97	5,820	2891–2929	50.33	50.0	5820–5820	0.0	97–97
98	5,880	2940–2940	50.00	50.0	5880–5880	0.0	98–99
99	5,940	2952–2988	50.30	50.0	5940–5940	0.0	99–99
100	6,000	3000–3000	50.00	50.0	6000–6000	0.0	100–100

Appendix F Datasheet values used

Table 49. Datasheet parameters used in the calculations [D].

Part	Parameter	Value	Used in
TCRT5000	V_F at I_F = 60 mA	1.25 V typ, 1.5 V max	Eq. 2
TCRT5000	I_F absolute maximum	60 mA	4.2
TCRT5000	I_C at V_{CE} = 5 V, I_F = 10 mA, D = 12 mm	0.5 / 1 / 2.1 mA	4.2
TCRT5000	$V_{CE(sat)}$ at I_F = 10 mA, I_C = 0.1 mA	≤ 0.4 V	4.2
TCRT5000	operating range (> 20 % relative I_C); peak distance	0.2–15 mm; 2.5 mm	3.4
CD4093B	V_P, V_N, V_H at 5 V and 10 V (inputs tied)	Table 6	4.2
CD4093B / CD4026B	I_{OH} at V_{DD} = 5 V, V_O = 4.6 V	−0.51 mA min, −1 mA typ	4.5, 4.8
CD4093B / CD4026B	I_{OH} at V_{DD} = 5 V, V_O = 2.5 V	−1.6 mA min, −3.2 mA typ	4.5, 4.8
CD4026B	V_{IH} min at V_{DD} = 5 V / 10 V	3.5 V / 7 V	4.5, 4.7
CD4026B	reset pulse width min at 5 V	200 ns	4.5
CD4017B	carry out	high for counts 0–4; one cycle per 10 clocks	4.6
CD4017B	clock frequency max at 5 V	2.5 MHz	4.6
NE555	trigger / threshold levels	$V_{CC}/3$ / $2V_{CC}/3$	4.6
NE555	supply current at 5 V, no load	3 mA typ (output low), 2 mA typ (output high); 6 / 5 mA max	5.8
NE555	V_{OH} at V_{CC} = 5 V, I_{OH} = −100 mA	2.75 V min, 3.3 V typ	4.6
PC817	CTR at I_F = 5 mA, V_{CE} = 5 V	50–400 %	4.6

Operating-point method. For a CMOS output driving a resistive load, the output source current was taken as a straight line through the two datasheet points (V_O = 4.6 V and 2.5 V at V_{DD} = 5 V, typical or minimum column) and intersected with the load line. Results: reset network 1 kΩ: 2.84 V typ / 1.91 V min; 10 kΩ: 5.0 V typ (limited by the supply) / 4.68 V min; D4 with 1 kΩ and V_F = 2.0 V: 1.82 mA typ / 1.22 mA min; segment with 220 Ω and V_F = 2.0 V: 3.03 mA typ / 1.67 mA min; set-pulse LED with 1 kΩ and V_F = 1.2 V: 2.23 mA typ / 1.50 mA min [C].

Appendix G Project timeline

Table 50. Timeline [F: project log].

Date (2026)	Milestone
16 Sep	Schematic complete (CD4026 variant, gate option 3); PCB placed and autorouted (Freerouting passes r1–r4); segment resistors 470 → 220 Ω; 5 mm corner radius; four M3 holes; zones filled; reference designators hidden; 65 value labels placed by solver; back-side artwork
17 Sep	Silkscreen revised by hand (mode table, 220R group labels, minimum text 0.8 mm); fabrication package regenerated, DRC 0 errors, JLCPCB check READY; accuracy analysis and behavioural simulation
22 Sep	Builder's inventory applied; fitted-value substitutions (R1, R10, R15, R16); purchase list by shop
23 Sep	3D models attached to all 95 parts (copper verified unchanged); schematic redrawn with wires only (netlist identical); fitted-parts assembly map
24 Sep	Board 1 assembled; display stuck at 0, no IR emission
25 Sep	Gerber-level connectivity check (149/149 nets, 372/372 pins); TCRT5000 LED polarity identified and fixed; counter mode working
26 Sep	Board 1 damaged by a supply short; board 2 built stage by stage; working; RPM readings close to a Lutron DT-2236
27 Sep	This report: data re-extracted from the design files, ERC/DRC re-run, schematic-to-PCB connectivity re-proven

Appendix H Abbreviations

Table 51. Abbreviations.

Term	Meaning	Term	Meaning
BOM	bill of materials	NPTH / PTH	non-plated / plated through-hole
CC	common cathode	RPM / RPS	revolutions per minute / second
CTR	current transfer ratio (optocoupler)	SES / DSN	Specctra session / design file
DRC / ERC	design / electrical rule check	THT	through-hole technology
IPC	inter-process communication (KiCad API)	V_{IH} / V_{IL}	input high / low voltage
N_m	number of reflective marks on the rotor	V_P / V_N	positive / negative Schmitt thresholds